

New insights into error accumulation due to biased particle distribution in semi-implicit particle methods

Guangtao Duan^{a,*}, Takuya Matsunaga^b, Seiichi Koshizuka^b, Akira Yamaguchi^{a,c},
Mikio Sakai^{a,d,**}

^a Department of Nuclear Engineering and Management, The University of Tokyo, 7-3-1, Hongo, Bunkyo-ku, Tokyo 113-8656, Japan

^b Department of Systems Innovation, The University of Tokyo, 7-3-1, Hongo, Bunkyo-ku, Tokyo 113-8656, Japan

^c Nuclear Professional School, School of Engineering, The University of Tokyo, 2-22 Shirakata, Tokai-mura, Naka-gun, Ibaraki 319-1188, Japan

^d Resilience Engineering Research Center, The University of Tokyo, 7-3-1, Hongo, Bunkyo-ku, Tokyo 113-8656, Japan

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Abstract

This study investigates the instability issue at a free surface when the consistent schemes based on variable differences are applied in semi-implicit particle methods. A semi-analytical error-analysis method is proposed to clarify how the incomplete/biased neighbor support triggers error accumulation and instability. Specifically, the discretization models are decomposed into the center-variable components (CVCs) and neighbor-variable components (NVCs). The influence of different components on error accumulation is analyzed theoretically and numerically. Based on the error analysis, new indices are proposed to evaluate the risk of error accumulation due to the biased neighbor support. Then, novel free-surface-detection conditions are proposed from the indices by detecting the particles prone to error accumulation as free surface particles. Numerical examples demonstrated that the proposed conditions detected fewer free-surface particles but produced more stable simulations compared to the existing conditions.

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1. Introduction

Mesh-free particle methods have great potential for simulating free-surface flows due to their intrinsically Lagrangian nature. Smoothed particle hydrodynamics (SPH) [1,2] and moving particle semi-implicit (MPS) [3,4] methods are well-known particle methods. The study on error analysis plays an important role in improving both stability and accuracy.

The common discretization models in particle methods can be roughly clarified into two groups. The first group is the well-known conservative models that are widely adopted in (weakly) compressible SPH [5]. For example, the pressure gradient model in this group is usually based on the pressure summation (or the like). This group can produce stable results and has become the mainstream. Nevertheless, it is difficult and time-consuming to

* Corresponding author.

** Corresponding author at: Resilience Engineering Research Center, The University of Tokyo, 7-3-1, Hongo, Bunkyo-ku, Tokyo 113-8656, Japan.

E-mail addresses: guangtao_duan@dem.t.u-tokyo.ac.jp (G. Duan), mikio_sakai@n.t.u-tokyo.ac.jp (M. Sakai).

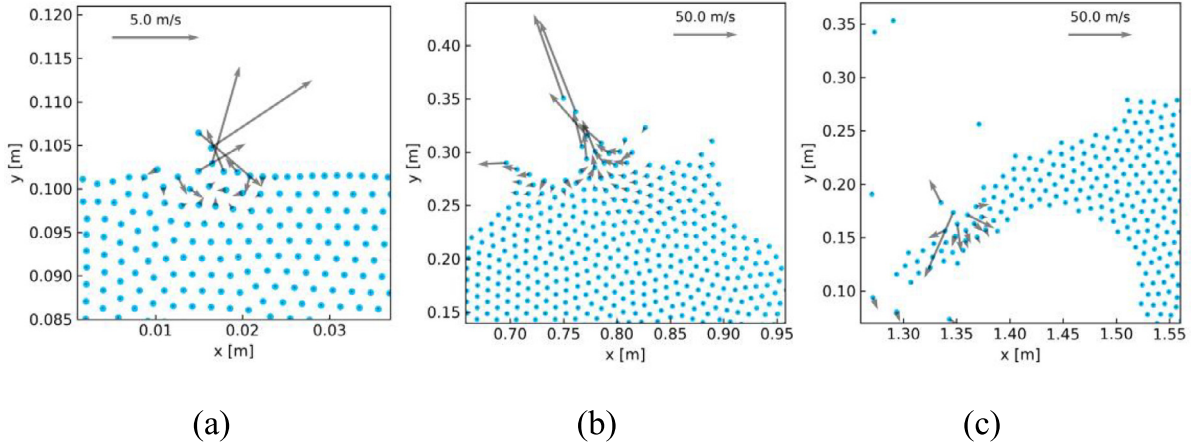


Fig. 1. Instability examples when the consistent schemes are directly applied at free surface. The arrows indicate velocity.

exactly restore the consistency for the conservative models [6]. The second group is the variable-difference-based models. For example, the gradient model is usually based on relative pressure. One advantage is the easy extension to the consistent high-order schemes by the kernel–gradient correction in SPH [7,8], least squares MPS methods (LSMPS) [9–11], the corrective matrix in MPS [12–14], and so on. However, instability is the main problem [5]. For example, the consistent schemes do reduce the error on the static particle distributions, but incredibly their dynamic error in simulations might increase fast and trigger instability. Therefore, if appropriate stabilization strategies can be developed, the second group may produce accurate and high-quality results [10,11]. The instability problem of the second group is investigated in the study. A corrected MPS method with the consistent schemes and the semi-implicit algorithm [15] is employed.

The particle shifting (PS) technique proposed by Xu et al. [16] is effective to enhance the stability of the second group, especially for internal particles. For example, Sun et al. [17] developed a tensile instability control (TIC) technique, where the conservative pressure model was employed near a free surface and the pressure-difference-based gradient model was applied under free surfaces in SPH. In this manner, the tensile instability was effectively suppressed, and accuracy was improved. Duan et al. [14] also developed such an idea in the MPS framework. Moreover, this idea has been adopted in many studies [13,14,17–20] to improve accuracy of internal particles. However, if the consistent schemes are directly applied near a free surface, instability can still happen easily [11,15], even though more advanced PS techniques specially designed for free surfaces [14,21,22] are implemented. For example, Fig. 1 presents three instability cases when the consistent high-order schemes [15] and the advanced PS methods [14,22] are directly applied near free surfaces. We observed that the spurious currents grew quickly in a few time steps. Namely, fast error accumulation took place and triggered instability. At first glance, the velocity fluctuations are basically random, and thus it is difficult to identify the intrinsic reason. In this situation, an in-depth error analysis becomes indispensable to suppress the spurious currents.

In explicit SPH methods, several instabilities, such as zero-energy modes [23], tensile instability [24], and pairing instability [25], have been carefully analyzed. The zero-energy modes usually take place in the simulations of solid mechanics. The tensile instability is mainly due to negative pressure, but the accurate consistent schemes can effectively suppress it [12,17,26]. Thus, the observed instability is not tensile instability. Pairing instability is caused by misusing kernels. Specifically, when the interparticle distance became small, the particles would not gain enough repulsive forces from the neighbors [5,27]. However, this problem does not take place in MPS, because the interaction forces always increase dramatically as particles approach [12]. Hence, the observed instability is not pairing instability. Price [5] once mentioned the instability (exponential growth of error) of the second-group models, which is similar to the observed ones in Fig. 1. However, the reason was simply attributed to the lack of conservativity and the error due to random particle distributions [5]. Besides, Khayyer and Gotoh [12] argued that the MPS method seemed more prone to (tensile) instability than SPH based on the pair-wise particle interactions. However, the concept of pair-wise interactions immediately becomes inapplicable when the corrective matrix is

adopted. Briefly, the above stability analysis was mainly based on the explicit algorithm [5,23–25] and the original zero-order models [5,12,23–25]. To our best knowledge, the stability analysis for the particle method using consistent schemes and a semi-implicit algorithm is still lacking. A numerical method is said to be stable if it does not magnify the errors that appear during numerical solution process [28]. In this viewpoint, the instability considered in this study is essentially the fast growth or accumulation of error. We will use error analysis to investigate how the error grows rapidly to trigger the instability in Fig. 1.

Based on our experiences, the instability in Fig. 1 is due to the mistaken detection of free-surface particles. Various detection methods, such as the particle number density (PND) [3], the neighboring-particle number [29], and the divergence of the position vector [30], the arc method [31], the parachute-scanning method [32], and the illumination method [33], have been developed in literature. We tested some combinations of the detection methods in Ref. [3,9,29,30,32] and found that the adjustable detection parameters could affect the occurrence time but could not thoroughly suppress the instability. Meanwhile, the compensation of virtual particles outside a free surface can improve the stability [14,33–39]. Particularly, Shibata et al. [33] precisely considered the virtual-particle contribution and derived new MPS models. The method could stably simulate various free surface flows [33]. However, the method was based on the zero-order MPS models rather than the consistent schemes. Recently, Matsunaga et al. [11] proposed a moving surface mesh for the free surface boundary based on LSMPS. Accuracy and stability were remarkably enhanced. However, the extension to 3D and the treatment of various topological changes of free surfaces were rather challenging and difficult. Because the instability was closely related to the free-surface treatments, the incomplete/biased neighbor support seemed the main reason. Even though it was well known that the anisotropic/biased particle distributions could easily trigger instability, the reason was still not thoroughly investigated. In this situation, the two related objectives of this study are as follows:

- (a) To clarify how biased particle distributions trigger instability for consistent schemes.
- (b) To propose a new free-surface-detection technique to stabilize the simulations.

The outline of this paper is as follows. The numerical method is presented in Section 2. A new semi-analytical error analysis method is proposed to investigate spurious currents at free surfaces in Section 3. Based on the error analysis, new indices are proposed for the free-surface detection in Section 4. Lastly, various numerical examples are performed to test the stability performance in Section 5.

2. Numerical methods

The mass and momentum conservation equations for incompressible flows in a Lagrangian framework are

$$\frac{D\rho}{Dt} = -\rho \nabla \cdot \mathbf{u} = 0, \quad (1)$$

and

$$\frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho} \nabla p + \frac{\mu}{\rho} \nabla^2 \mathbf{u} + \mathbf{g}, \quad (2)$$

where $\mathbf{u}$ is the velocity, p is the pressure, $\mathbf{g}$ is gravity, and ρ and μ are the density and dynamic viscosity, respectively.

2.1. Consistent MPS discretization models with corrective matrix

The MPS models with a second-order corrective matrix (SCM) were derived from the least squares method in Ref. [13]. A precise coupling of the Dirichlet, Neumann and Robin boundary conditions into the discretization models was possible, as developed in Ref. [10,15,40]. We adopt these schemes from Ref. [15] in this study.

For an internal fluid particle i , the neighboring particle sets and the roles in the pressure boundary conditions are sketched in Fig. 2. Specifically, the set N_S contains all the neighboring particles, and it is used for calculating the stabilization terms (e.g., PS and PND). The set N_D includes the fluid, free-surface, and wall particles in the neighborhood for the physical discretization. For a center particle i , the neighboring particles j are classified into three types based on the (pressure) boundary conditions: internal, Dirichlet, and Neumann, as shown in Fig. 2. When the Neumann boundary condition is enforced on a wall particle, the boundary normal vector $\mathbf{n} = (n_x, n_y)$ is given, and the normal boundary flux $\partial p / \partial n$ can be calculated from [10]:

$$\frac{\partial p}{\partial n} = \rho \mathbf{n} \cdot \mathbf{g}. \quad (3)$$

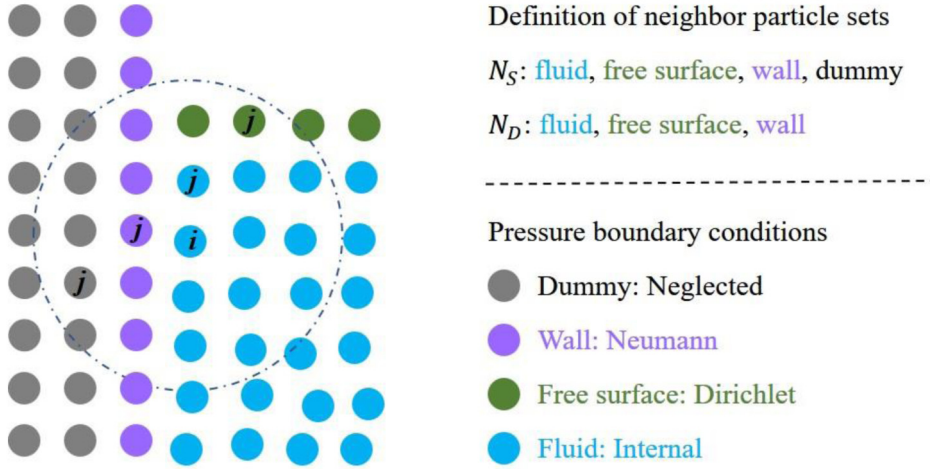


Fig. 2. Sketch of various neighboring-particle sets and pressure boundary-condition types for an internal fluid particle i .

The corrective matrix with the boundary conditions coupled is first described below. A general directional vector $\mathbf{P}$ is defined between the reference particle i and a neighboring particle j as follows [15]:

$$\mathbf{P} = \begin{cases} \begin{bmatrix} \frac{x_{ij}}{r_{ij}} & \frac{y_{ij}}{r_{ij}} & \frac{x_{ij}^2}{l_0 r_{ij}} & \frac{y_{ij}^2}{l_0 r_{ij}} & \frac{x_{ij} y_{ij}}{l_0 r_{ij}} \end{bmatrix}^T, & j \in \text{Internal or Dirichlet} \\ \begin{bmatrix} n_x & n_y & \frac{2n_x x_{ij}}{l_0} & \frac{2n_y y_{ij}}{l_0} & \frac{n_x y_{ij} + n_y x_{ij}}{l_0} \end{bmatrix}^T, & j \in \text{Neumann} \end{cases}, \quad (4)$$

where $\mathbf{r}_i = (x_i, y_i)$ is the position vector of particle i ; $x_{ij} = x_j - x_i$, $y_{ij} = y_j - y_i$, $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$, $r_{ij} = \|\mathbf{r}_{ij}\|$, l_0 is the constant initial particle distance, and $\mathbf{n} = (n_x, n_y)$ is the boundary normal vector at particle j . Here, $j \in \text{Internal or Dirichlet}$ means that j is an internal particle or a Dirichlet boundary particle (see Fig. 2), and $j \in \text{Neumann}$ means that j is a Neumann boundary particle (see Fig. 2). For convenience, the vector $\mathbf{P}$ is rewritten in the following general form:

$$\mathbf{P} = [P_1 \ P_2 \ P_3 \ P_4 \ P_5]^T. \quad (5)$$

The moment matrix $\mathbf{M}$ is calculated from $\mathbf{P}$ as follows:

$$\mathbf{M} = \begin{bmatrix} (P_1, P_1) & (P_1, P_2) & \cdots & (P_1, P_5) \\ (P_2, P_1) & (P_2, P_2) & \cdots & (P_2, P_5) \\ \vdots & \vdots & \ddots & \vdots \\ (P_5, P_1) & (P_5, P_2) & \cdots & (P_5, P_5) \end{bmatrix}, \quad (6)$$

where (P_α, P_β) is the inner product of the components P_α and P_β obtained by looping through all the neighboring particles of the set N_D , using the dimensionless weights w_{ij}/n_0 , as follows:

$$(P_\alpha, P_\beta) = \sum_{j \in N_D} P_\alpha \cdot P_\beta \frac{w_{ij}}{n_0}. \quad (7)$$

In Eq. (7), w_{ij} is the weight function between particle j and i , and n_0 is the constant PND. The adopted weight function is as follows:

$$w_{ij} = w(r_{ij}, r_e) = \begin{cases} \left(\frac{r_{ij}}{r_e} - 1 \right)^2 & r_{ij} < r_e \\ 0 & r_{ij} \geq r_e \end{cases}, \quad (8)$$

where r_e is the particle-interaction radius. After $\mathbf{M}$ is calculated, the corrective matrix $\mathbf{C}$ is just the inverse of $\mathbf{M}$, namely $\mathbf{C} = \mathbf{M}^{-1}$. Note that $\mathbf{P}$, $\mathbf{C}$, and $\mathbf{M}$ are all dimensionless. As pointed out by Tamai and Koshizuka [9], the condition number of $\mathbf{M}$ is greatly reduced due to the dimensionless feature.

Next, we describe the consistent discretization models. After the corrective matrix is considered, the high-order gradient, divergence, and Laplacian models are still similar to the original ones [13]. Specifically, the high-order gradient model becomes:

$$\langle \nabla \phi \rangle_i = \frac{1}{n_0} \sum_{j \in N_D} \left\{ w_{ij} d_{ij} \left(\begin{bmatrix} \mathbf{C}_1 \\ \mathbf{C}_2 \end{bmatrix} \mathbf{P} \right) \right\}, \quad (9)$$

where $\mathbf{C}_1$ and $\mathbf{C}_2$ are the first and second rows of matrix $\mathbf{C}$, respectively. The directional difference scalar d_{ij} in Eq. (9) is defined as follows:

$$d_{ij} = \begin{cases} \frac{\phi_j - \phi_i}{r_{ij}}, & j \in \text{Internal or Dirichlet} \\ \frac{\partial \phi_j}{\partial n}, & j \in \text{Neumann} \end{cases}, \quad (10)$$

where ϕ is a scalar variable (e.g., pressure p). The dimensionless product of $[\mathbf{C}_1 \ \mathbf{C}_2]^T$ and $\mathbf{P}$ corresponds to the direction vector between two particles (i.e., $\mathbf{r}_{ij}/r_{ij}$). Similarly, the high-order divergence model becomes

$$\langle \nabla \cdot \mathbf{u} \rangle_i = \frac{1}{n_0} \sum_{j \in N_D} \left\{ w_{ij} \mathbf{d}_{ij} \cdot \left(\begin{bmatrix} \mathbf{C}_1 \\ \mathbf{C}_2 \end{bmatrix} \mathbf{P} \right) \right\}, \quad (11)$$

where the directional difference vector $\mathbf{d}_{ij}$ is defined as follows:

$$\mathbf{d}_{ij} = \begin{cases} \frac{\mathbf{u}_j - \mathbf{u}_i}{r_{ij}}, & j \in \text{Internal or Dirichlet} \\ \frac{\partial \mathbf{u}_j}{\partial n}, & j \in \text{Neumann} \end{cases}. \quad (12)$$

Accordingly, the high-order Laplacian model is given by

$$\langle \nabla^2 \phi \rangle_i = \frac{2}{n_0} \sum_{j \in N_D} \left\{ w_{ij} d_{ij} \frac{[\mathbf{C}_3 + \mathbf{C}_4] \mathbf{P}}{l_0} \right\}, \quad (13)$$

where $\mathbf{C}_3$ and $\mathbf{C}_4$ are the third and fourth rows of $\mathbf{C}$, respectively; the directional difference d_{ij} is equal to the gradient, as defined in Eq. (10). Note that the product of $[\mathbf{C}_3 + \mathbf{C}_4]$ and $\mathbf{P}$ in Eq. (13) is a dimensionless scalar.

In Refs. [13,15], the convergence was demonstrated on the complete neighbor support [Fig. 3(a)]. The convergence behavior on the incomplete/biased neighbor support [Fig. 3(b)] is discussed below because the schemes are directly applied at free surface. The adopted test function from Ref. [14] is:

$$\phi(x, y) = 5x + 3y + 10x^2 + 30xy + 20y^2 + 15x^3 + 30x^2y + 40xy^2 + 25y^3. \quad (14)$$

The biased particle distribution with fluctuations is generated below. First, a uniform Cartesian particle distribution is produced with l_0 . Second, random fluctuations up to $0.1 \times l_0$ are added to particle positions. Third, when a particle's angle going counterclockwise from the positive x -axis is less than 216° , the particles will be deleted, as shown in Fig. 3(b). The convergence results on the complete and biased neighbor supports are compared in Fig. 4. The convergence could be observed, demonstrating the consistency. The gradient accuracy was basically not influenced. However, the Laplacian accuracy was significantly reduced, because the error in different directions could not be well cancelled out. These error behaviors were consistent with the analysis in Ref. [13]. Even though the schemes are consistent at free surfaces, fast error accumulation takes place easily. This mysterious phenomenon becomes the subject of this study.

2.2. Special modification to a singular moment matrix near a free surface

Near free surfaces, a singular moment matrix may take place, and special treatment is necessary. Specifically, when the number of neighboring particles in N_D is less than the rank of $\mathbf{M}$ (5 in 2D), $\mathbf{M}$ will be ill-conditioned.

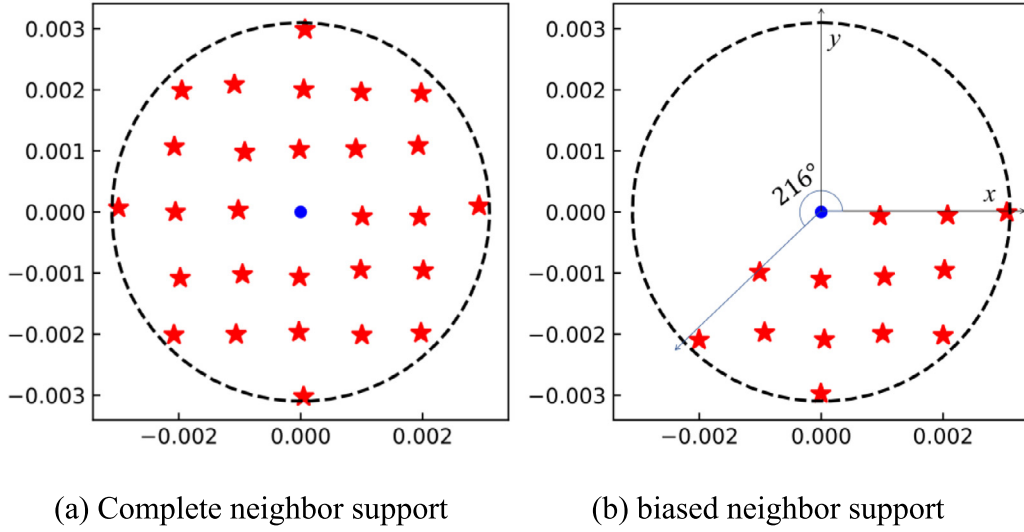


Fig. 3. Sketch of the complete and biased neighbor supports for the convergence tests.

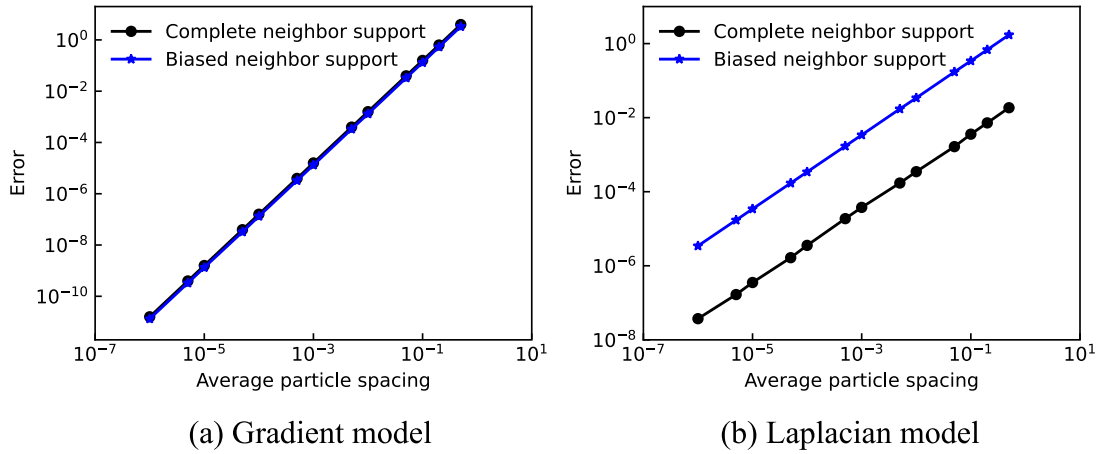


Fig. 4. Convergence comparison of the gradient and Laplacian models based on the complete and biased neighbor supports.

Inspired by Zhang & Liu [41], when the particle number in the set N_D is less than 7 in 2D (the critical number 7 rather than 5 is selected for a possible sufficient condition), the following modification is made to $\mathbf{M}$ in this study: *all the off-diagonal elements of $\mathbf{M}$ are reset to zero*. In this case, $\mathbf{M}$ becomes an invertible diagonal matrix, and all the derivatives along a specific axis are approximated using information on that axis only. The resulting schemes are similar to those in Refs. [41,42]. It is noted that the scheme accuracy will be reduced when this modification is applied.

2.3. Existing free-surface detection methods

An advanced method for free-surface detection is to scan a parachute region as proposed by Marrone et al. [32]. A modified version is adopted here as an example of existing methods.

The detection conditions are like those in Refs. [9,36]. The surface normal of particle i is computed from

$$\mathbf{n}_i = \frac{N_i}{\|N_i\|}, \text{ where } N_i = \frac{1}{n_i} \sum_{j \in N_S} \left\{ \frac{\mathbf{r}_i - \mathbf{r}_j}{r_{ij}} w_{ij} \right\}. \quad (15)$$

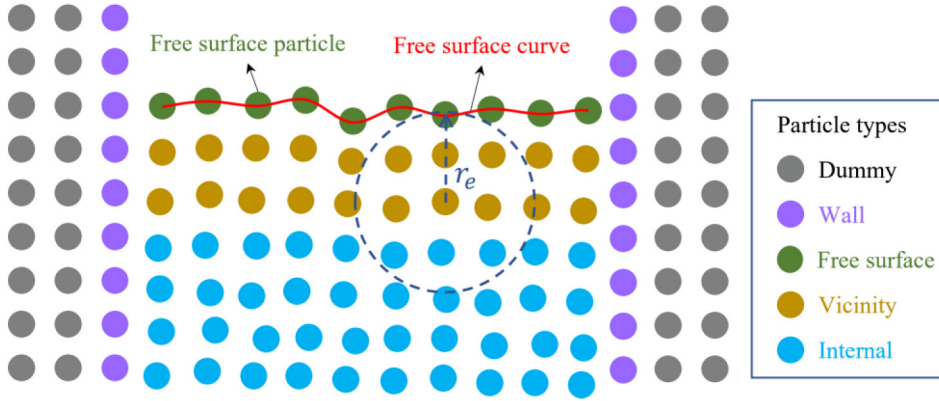


Fig. 5. Sketch for the particle clarification and the boundary fluctuation at a free surface.

The following conditions are used for the free-surface detection:

$$\begin{cases} \text{if } (n_i < 0.97n_0 \text{ or } N_i < 0.85N_0 \text{ or } \|N_i\| > 0.05) & \text{Parachute region detection} \\ \text{else} & \text{Internal particle} \end{cases}, \quad (16)$$

where n_i is the PND, N_i is the number of neighboring particles, and N_0 is the initial constant number of neighboring particles. The “parachute-region-detection” tests whether a neighboring particle exists in the parachute region, based on the following conditions:

$$\begin{cases} \|r_j - r_i\| \geq \sqrt{2}l_0 \\ \|(r_i + l_0 m_i) - r_j\| < l_0 \end{cases}, \quad (17)$$

or

$$\begin{cases} \|r_j - r_i\| < \sqrt{2}l_0 \\ \frac{r_j - r_i}{\|r_j - r_i\|} \cdot m_i > \cos\left(\frac{\pi}{4}\right) \end{cases}. \quad (18)$$

If Eq. (17) or (18) is satisfied, particle i is detected as an internal particle; otherwise, particle i is a free-surface particle. These existing conditions could produce stable simulations when the conservative pressure gradient model (of the first group) was applied near free surfaces [13–15,18,36,43–45].

2.4. Stabilization strategy

The PS techniques for keeping the particles evenly spaced are described in this section. After free surface particles are detected, the (free-surface) vicinity particles are defined as the particles whose closest distance to a free surface particle is smaller than r_e . The clarification of free surface, vicinity and internal particles are shown in Fig. 5.

First, PS is applied to all the internal particles. Specifically, the shifting vectors are calculated from [13]:

$$\delta r_i = \frac{0.15l_0}{n_0} \sum_{j \in N_S} \left\{ \frac{l_0}{r_{ij}} \frac{r_{ij}}{r_{ij}} w_{ij} \right\}. \quad (19)$$

Second, the OPS method [21] is modified and applied to the free surface particles:

$$\delta r'_i = (\mathbf{I} - \mathbf{n}_i \otimes \mathbf{n}_i) \delta r_i, \quad (20)$$

where $\mathbf{I}$ is the identity matrix, and $\mathbf{n}_i$ is the surface normal calculated from Eq. (15), similar to [14]. Third, IPST [22] is applied to the vicinity particles by

$$\delta r''_i = \frac{0.15l_0}{n_0} \sum_{(j \in N_S \text{ and } r_{ij} < (r_f + 0.1l_0))} \left\{ \frac{l_0}{r_{ij}} \frac{r_{ij}}{r_{ij}} w_{ij} \right\}, \quad (21)$$

where r_f is the distance to the closest free-surface particle.

One problem of the detected free surface is that the fluctuations will inevitably happen as shown in Fig. 5. Such fluctuations result in small velocity/pressure fluctuations near free surface in high-order schemes, as stressed in [11,15]. To suppress the fluctuations, slight artificial viscosity in SPH [46] is adopted only for the free-surface and vicinity particles:

$$\Delta \mathbf{u}_i = -0.05 \frac{d\Delta t}{n^0} \sum_{j \in N_S} \Pi_{ij} \frac{\mathbf{r}_{ij}}{r_{ij}} \frac{w_{ij}}{r_{ij}}, \quad (22)$$

where the term Π_{ij} is given by

$$\Pi_{ij} = \begin{cases} -c_0 u_{ij} + 2u_{ij}^2 & \mathbf{u}_{ij} \cdot \mathbf{r}_{ij} < 0 \\ 0 & \mathbf{u}_{ij} \cdot \mathbf{r}_{ij} > 0 \end{cases}, \quad (23)$$

In Eq. (23), the relative velocity of approach is calculated from

$$u_{ij} = \frac{h \mathbf{u}_{ij} \cdot \mathbf{r}_{ij}}{r_{ij}^2 + 0.01h^2}, \quad (24)$$

where $h = 0.5r_e$, and the sound speed is $c_0 = 10U_{max}$. It must be noted that the artificial viscosity can only suppress the small fluctuations in Fig. 5 but cannot prevent the fast-error-accumulation instability in Fig. 1.

2.5. Overall time marching algorithm

A semi-implicit algorithm is used for the time integration. Similar to Ref. [15], the solution algorithm and enforced boundary conditions are as follows.

(1) The temporary position is calculated for all the fluid particles:

$$\mathbf{r}^* = \mathbf{r}^k + \mathbf{u}^k \Delta t. \quad (25)$$

(2) The corrective matrix is calculated and stored. Meanwhile, the free-surface, vicinity, and internal particles are detected.

(3) The velocity divergence for the PPE source is calculated from Eq. (11) for the internal and vicinity particles.

(3) The following PPE is solved for the internal and vicinity particles:

$$\frac{1}{\rho} \langle \nabla^2 p \rangle^{k+1} = (1 - \gamma) \frac{1}{\Delta t} \langle \nabla \cdot \mathbf{u} \rangle^k - \gamma \frac{1}{\Delta t^2} \left\{ \frac{n^* - n_0}{n_0} \right\}, \quad (26)$$

where γ (0.001 by default) is the coefficient for the mixing PPE source [47], and n^* is the temporary PND. A constant reference pressure ($p_{ref} = 0$ by default) is imposed to free surface particles. The Neumann boundary condition in Eq. (3) is imposed to the wall particles.

(4) The gravity acceleration is computed for all the fluid particles:

$$\mathbf{u}^* = \mathbf{u}^k + \mathbf{g} \Delta t. \quad (27)$$

(5) The pressure gradient is calculated for all the fluid particles:

$$\mathbf{u}^{**} = \mathbf{u}^* - \frac{\Delta t}{\rho} \langle \nabla p \rangle^{k+1}. \quad (28)$$

(6) The final velocity is obtained by considering the viscosity term implicitly [18,48] for all the fluid particles:

$$\frac{\mathbf{u}^{k+1}}{\Delta t} - \frac{\mu}{\rho} \langle \nabla^2 \mathbf{u} \rangle^{k+1} = \frac{\mathbf{u}^{**}}{\Delta t}. \quad (29)$$

The no-slip boundary condition is considered at the wall particles. An approximate free-slip boundary condition, namely $\partial \mathbf{u} / \partial n = \mathbf{0}$ in Eq. (12), is applied for the free surface particles based on the constraint least squares

Table 1

The employed discretization and stabilization models for different types of particles.

Discretization/stabilization models	Free-surface particles	Vicinity particles	Internal particles
$\langle \nabla \cdot \mathbf{u} \rangle^k$ in Eq. (26)	Not applicable		Eq. (11)
$\langle \nabla^2 p \rangle^{k+1}$ in Eq. (26)	$p_i = p_{ref}$		Eq. (13)
$\langle \nabla p \rangle^{k+1}$ in Eq. (28)	Eq. (9)		Eq. (9)
$\langle \nabla^2 \mathbf{u} \rangle^{k+1}$ in Eq. (29)	Eq. (13)		Eq. (13)
Stabilization terms	Eqs. (20) and (22)	Eqs. (21) and (22)	Eq. (19)

method [9]. A more accurate treatment method for the free-slip boundary condition can be found in Ref. [11]. Note that the low viscosity in this study almost has no influence on the considered error accumulation.

(7) The final particle position for all the fluid particles is updated by:

$$\mathbf{r}^{k+1} = \mathbf{r}^k + \left(\frac{\mathbf{u}^k + \mathbf{u}^{k+1}}{2} \right) \Delta t. \quad (30)$$

(8) Various PS techniques for the fluid particles in Section 2.4 are applied to maintain the stability.

Even though the above algorithm was specially designed for the highly-viscous fluids [15], its performance for the low-viscosity fluids was almost the same with the original algorithm [3] based on the tests in Ref. [15]. Lastly, the discretization and stabilization models applied to the free-surface, vicinity, and internal particles are summarized in Table 1.

3. A new semi-analytical analysis method for error accumulation

When the neighbor support becomes biased, error accumulation and transfer take place simultaneously. To specifically study error accumulation, we identified a basic error-accumulation pattern (BEAP), where only one particle continuously underwent severe error growth. The BEAP examples are shown in Figs. 1(c) and 6. It is possible to study the accumulation mechanism in a single BEAP because error accumulation was dominant and error transfer became secondary. However, the commonly encountered instabilities in Fig. 1(a) and (b) were difficult to analyze because error accumulation and transfer were comparative.

The reason for BEAP is analyzed in this section. First, the main features of a BEAP are discussed in Section 3.1. Namely, the center variables (i.e., velocity and pressure) were much larger than the neighboring ones. Second, to investigate the contribution of the largest center variables, the discretization models are decomposed into the center-variable components (CVCs) and the neighbor-variable components (NVCs) in Section 3.2. It was observed that error accumulation just took place in the direction of CVC rather than a random direction, indicating that CVC was the main accumulation reason. Further, based on the given direction and sign of CVCs in Fig. 6, it was found that the error accumulated via a positive feedback loop of only CVCs, once again indicating dominant role of CVCs. Third, the error accumulation is derived analytically only based on CVCs in Section 3.3. It is demonstrated that CVCs have a tendency to cause error accumulation. Next, the influence of NVCs is investigated numerically in Section 3.4. It was found that the NVCs usually attempted to suppress error accumulation. Last, the influence of the Laplacian models on the PPE solution is discussed in Section 3.5.

3.1. Typical process of a BEAP

A typical process of BEAP in the hydrostatic-pressure problem is presented in Fig. 6. Several internal particles – A, B, C, D, and E – all faced biased neighbor supports. However, only particle C suffered from error accumulation. Specifically, from Fig. 6(a) to (c), velocity increased continuously in a specific direction. At the 8th time step in Fig. 6(d), the internal particle C was finally detected as a free-surface particle, and the error accumulation stopped. Namely, detecting a particle as a free-surface particle could prevent error accumulation.

During error accumulation, the velocity and pressure fluctuations were tightly coupled, because their magnitude increased simultaneously in Fig. 6(a–c). Even though PPE was solved globally, the velocity/pressure profiles far

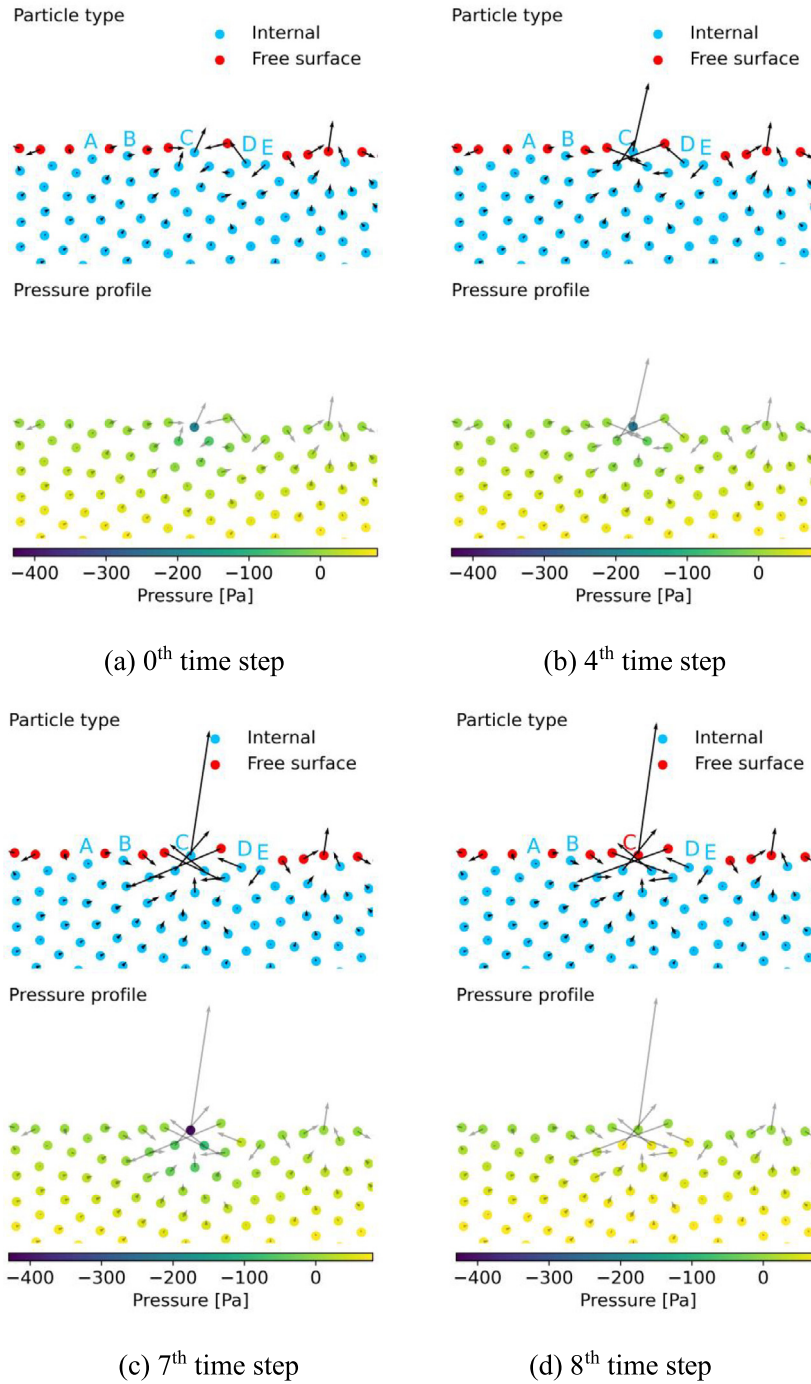


Fig. 6. A typical process of BEAP near a free surface. The upper part of each subfigure shows the particle types, the lower part shows pressure profiles, and the arrows indicate particle velocities.

from particle C were hardly affected, indicating that the error-transfer rate was smaller than the error-accumulation rate. Namely, the BEAP was still a local phenomenon. Because the velocity and pressure of particle C was locally largest, they certainly played a major role in BEAP. Therefore, the discretization models are decomposed into CVCs and NVCs below to evaluate the influence of center variables.

3.2. Decomposition of discretization models and qualitative error accumulation

The velocity error is defined as $\Delta \mathbf{u} = \mathbf{u} - \bar{\mathbf{u}}$, where $\mathbf{u}$ and $\bar{\mathbf{u}}$ are the simulated and theoretical velocity, respectively. One challenge for error analysis is that theoretical solutions are usually unknown, and thus an assumption must be made. The fast error accumulation is studied here, and it can result in large fluctuations (e.g., Figs. 1 and 6). In this situation, $\mathbf{u}$ is much larger than $\bar{\mathbf{u}}$, and thus the latter is assumed to be negligible. Briefly, errors and primitive variables can be regarded as the same during fast error accumulation. In this paper, the error accumulation is derived based on primitive variables, and thus the discretization models are directly decomposed.

First, the gradient model is considered for decomposition. When the Neumann boundary condition is not considered, the gradient model can be decomposed into:

$$\langle \nabla p \rangle_i = \frac{1}{n_0} \sum_{j \in N_D} \left\{ w_{ij} \frac{p_j}{r_{ij}} \left(\begin{bmatrix} C_1 \\ C_2 \end{bmatrix} \mathbf{P} \right) \right\} - \frac{p_i}{n_0} \sum_{j \in N_D} \left\{ w_{ij} \frac{1}{r_{ij}} \left(\begin{bmatrix} C_1 \\ C_2 \end{bmatrix} \mathbf{P} \right) \right\}. \quad (31)$$

Then Eq. (31) can be reformulated as

$$\langle \nabla p \rangle_i = \mathbf{G}_j + \mathbf{G}_i, \quad (32)$$

where the NVC due to p_j is:

$$\mathbf{G}_j = \frac{1}{n_0} \sum_{j \in N_D} \left\{ w_{ij} \frac{p_j}{r_{ij}} \left(\begin{bmatrix} C_1 \\ C_2 \end{bmatrix} \mathbf{P} \right) \right\}, \quad (33)$$

and the CVC due to p_i is:

$$\mathbf{G}_i = -\frac{p_i}{n_0} \sum_{j \in N_D} \left\{ w_{ij} \frac{1}{r_{ij}} \left(\begin{bmatrix} C_1 \\ C_2 \end{bmatrix} \mathbf{P} \right) \right\}. \quad (34)$$

Meanwhile, one can rewrite $\mathbf{G}_i$ as

$$\mathbf{G}_i = -\frac{p_i}{r_e} \mathbf{b}_i, \quad (35)$$

where $\mathbf{b}_i$ is defined from the gradient-model coefficients:

$$\mathbf{b}_i = \sum_{j \in N_D} \left\{ \frac{w_{ij}}{n_0} \frac{r_e}{r_{ij}} \left(\begin{bmatrix} C_1 \\ C_2 \end{bmatrix} \mathbf{P} \right) \right\}. \quad (36)$$

Note that $\mathbf{G}_i$ can be only in the same or opposite direction of $\mathbf{b}_i$. In normal circumstances, the above decomposition is meaningless; but it becomes useful when the center error is obviously larger than the neighboring ones (e.g., Fig. 6). For example, when particle i has a pressure error Δp_i , based on Eq. (35), the gradient error is

$$\mathbf{E}_{G,i} = -\frac{\Delta p_i}{r_e} \mathbf{b}_i. \quad (37)$$

This kind of decomposition is particularly designed for the fast error accumulation.

Similarly, the divergence model can be decomposed. Namely, it can be written as:

$$\langle \nabla \cdot \mathbf{u} \rangle_i = D_j + D_i, \quad (38)$$

where D_j is the NVC due to $\mathbf{u}_j$:

$$D_j = \frac{1}{n_0} \sum_{j \in N_D} \left\{ w_{ij} \frac{\mathbf{u}_j}{r_{ij}} \cdot \left(\begin{bmatrix} C_1 \\ C_2 \end{bmatrix} \mathbf{P} \right) \right\}, \quad (39)$$

and D_i is the CVC due to $\mathbf{u}_i$:

$$D_i = -\frac{\mathbf{u}_i}{r_e} \cdot \mathbf{b}_i. \quad (40)$$

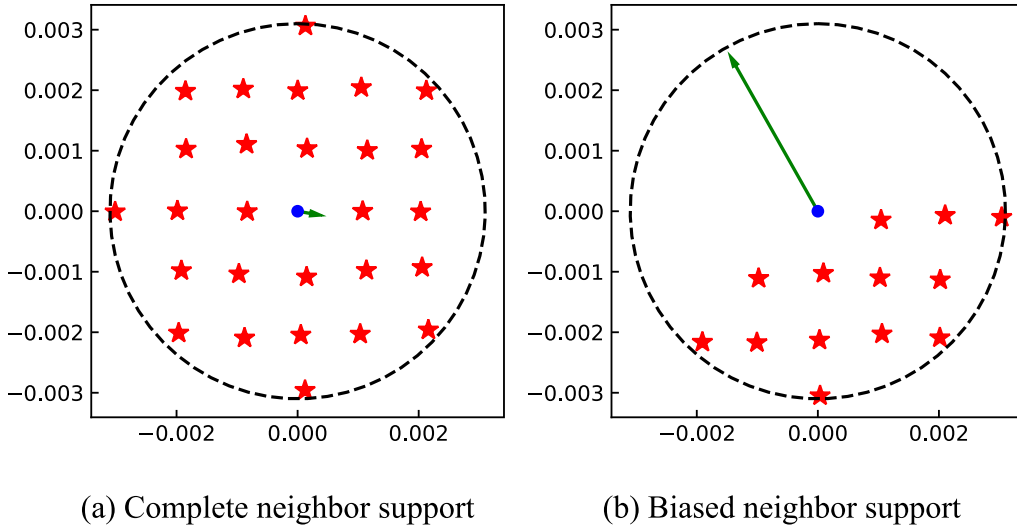


Fig. 7. The comparison of the gradient error produced by the same center fluctuation.

Note that $\mathbf{G}_i$ and D_i share the same $\mathbf{b}_i$. Similarly, the divergence error due to the error $\Delta \mathbf{u}_i$ is computed as:

$$E_{D,i} = -\frac{\Delta \mathbf{u}_i}{r_e} \cdot \mathbf{b}_i. \quad (41)$$

The center fluctuations are maximum in Fig. 6. Therefore, the gradient errors due to center fluctuations are compared on the complete and biased neighbor supports. Specifically, the particle distributions in Fig. 3 and the test function in Eq. (14) are employed. The same fluctuation $\Delta \phi_i = 0.001$ is enforced at the center particle. The calculated gradient errors are compared in Fig. 7. Obviously, the fluctuation could easily trigger larger gradient error when the neighbor support was biased. Particularly, this error is always in the direction of $\mathbf{b}_i$. Similarly, the divergence error can be large due to $\Delta \mathbf{u}_i$ for biased neighbor supports [see Eq. (41)]. Therefore, $E_{G,i}$ and $E_{D,i}$ are called the *bias errors*, and $\mathbf{b}_i$ is called a *bias-error index*.

Accordingly, the Laplacian model can be decomposed into:

$$\langle \nabla^2 p \rangle_i = L_j + L_i, \quad (42)$$

where L_j is the NVC due to p_j ,

$$L_j = \frac{2}{n_0} \sum_{j \in N_D} \left\{ w_{ij} \frac{p_j}{r_{ij}} \frac{[C_3 + C_4] \mathbf{P}}{l_0} \right\}, \quad (43)$$

and L_i is the CVC due to p_i ,

$$L_i = -\frac{p_i}{l_0 r_e} c_i. \quad (44)$$

In Eq. (44), the coefficient c_i is defined from the Laplacian-model coefficients:

$$c_i = 2 \sum_{j \in N_D} \left\{ \frac{w_{ij}}{n_0} \frac{r_e}{r_{ij}} ([C_3 + C_4] \mathbf{P}) \right\}. \quad (45)$$

where c_i is called a *constant-coefficient index* because it is around a constant for internal particles (see Fig. 8).

The influence of particle distributions on the pressure fluctuations caused by the PPE source is discussed below, considering that the Laplacian model is mainly used to solve PPE. The hydrostatic-pressure problem in Fig. 8 is considered. The PPE solution is based on the initial state with zero velocity. Thus, the PPE source is zero everywhere. The pressure profile is mainly produced by the Neumann boundary condition at bottom. The particle size is 0.005 m (see details in Section 5.1). An additional PPE-source fluctuation of 5000 1/s² is imposed to the treatment particles A–E in Fig. 8 one by one. This fluctuation corresponds to a positive velocity divergence. The

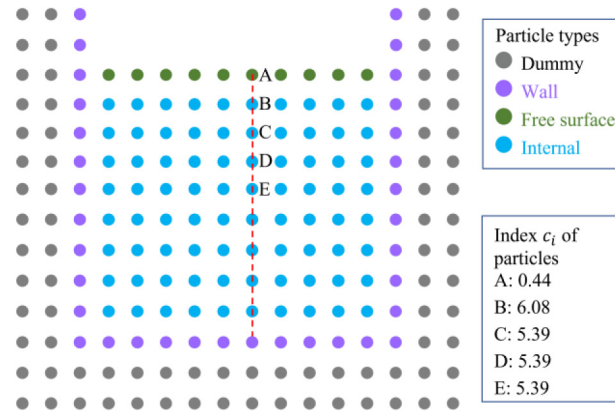


Fig. 8. Sketch for the treatment particles A–E in solving the PPE with source fluctuation in the hydrostatic-pressure problem.

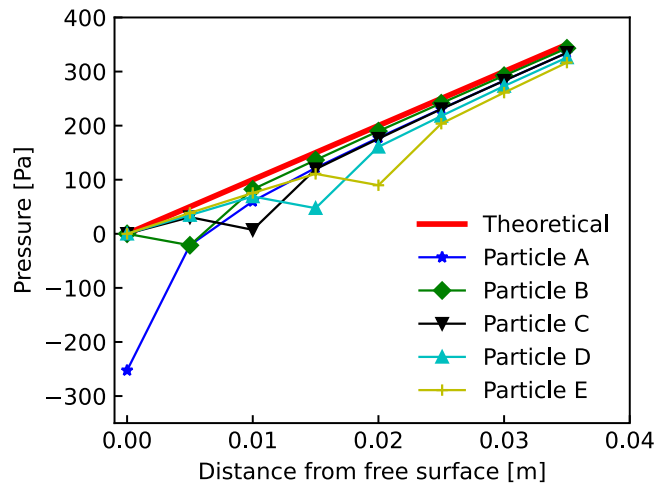


Fig. 9. Simulated pressure fluctuations when the same level of the PPE-source fluctuation was imposed to the treatment particle A, B, C, D, or E in Fig. 8. The imposed PPE-source fluctuation corresponds to a positive velocity divergence.

calculated pressure profiles along the vertical line in Fig. 8 are shown in Fig. 9. The lowest negative pressure always took place at the treatment particles. The fluctuated pressure of particle A was lowest. Namely, the same PPE-source fluctuation could generate much larger pressure errors when the neighbor support was biased. Briefly, the static results in Figs. 7 and 9 clearly demonstrated the consistent schemes are prone to large errors on biased particle distributions.

After the large errors due to fluctuations are confirmed on static biased particle distributions, the evolution direction of these errors is further discussed. The center pressure was always largest in Figs. 1(c) and 6, and its contribution to the pressure gradient could be calculated by Eq. (35). The error-accumulation direction and the $-\mathbf{G}_i$ direction in two cases are presented in Fig. 10, where the minus sign of $\mathbf{G}_i$ comes from the governing equation (2). One interesting phenomenon was that the two directions were almost the same, clearly indicating that the error accumulation was caused by $\mathbf{G}_i$. Namely, the NVC contribution was secondary. Furthermore, the accumulated velocity and pressure of particle C were given in Fig. 6, and thus it was possible to calculate D_i and $\mathbf{G}_i$. Specifically, based on D_i and $\mathbf{b}_i$ in Fig. 11, the outward velocity could always produce a positive velocity divergence. Based on Fig. 9, this velocity divergence could usually result in a negative pressure via PPE. This was consistent with the observed negative pressure in Fig. 6. Meanwhile, based on $\mathbf{G}_i$ and $\mathbf{b}_i$ in Fig. 11, the negative pressure could always

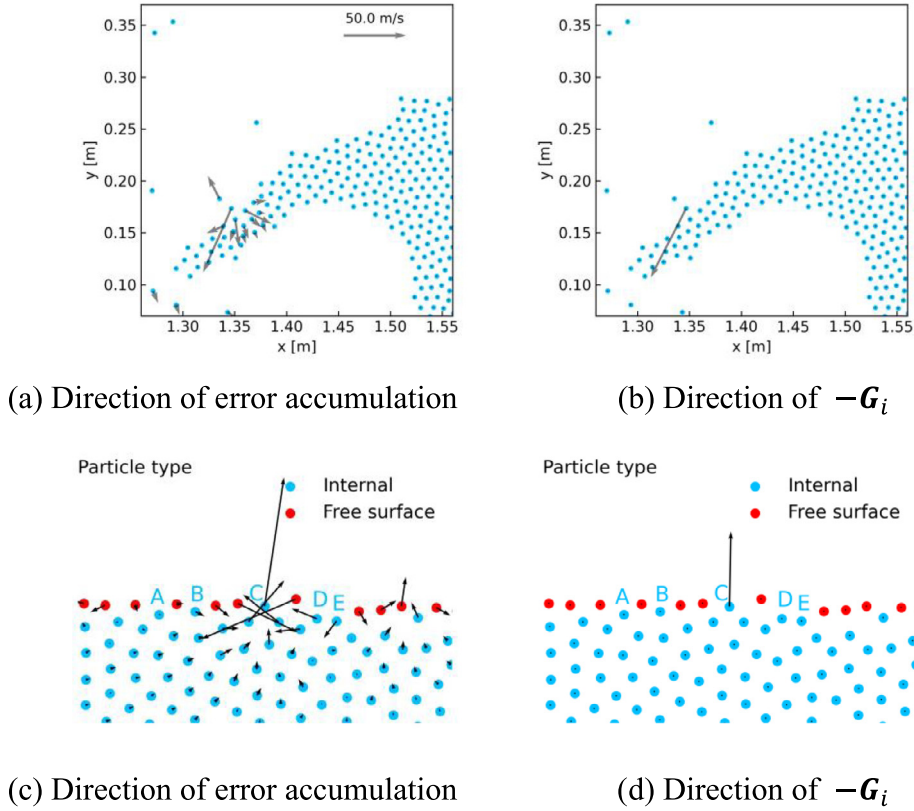


Fig. 10. Relation between the error-accumulation direction and the $-G_i$ direction in two cases. The upper case corresponds to Fig. 1(c). The lower case corresponds to Fig. 6.

produce an outward velocity, which just further enhanced the observed accumulative velocity. In brief, the pressure and velocity would excite each other just via the positive feedback loop of CVCs in Fig. 11 and thus accumulate simultaneously. This loop was obtained only from CVCs. Once again, it demonstrated that the CVCs were the main reason for error accumulation. The error evolution only based on CVCs is derived theoretically in Section 3.3. It seems that this loop still holds in an explicit algorithm, which requires further investigations in future.

3.3. Error accumulation caused by the center-variable components

The error accumulation is quite fast (just 7 time steps) in Fig. 6, the particle positions are hardly changed, and thus convection is not important. By contrast, the velocity divergence, PPE, and pressure gradient play a key role in error accumulation (see Fig. 11). When the dominant CVCs are only considered into the simplified governing equations, the error evolution can be derived. To reveal how the error grows rapidly and unphysically, the analysis aims to evaluate the error-growth ratio after just one time step $\Delta v_i^{k+1}/\Delta v_i^k$, where Δv_i^k is the magnitude of velocity error for particle i at the time step k . Based on the assumption that errors and primitive variables can be regarded as the same in Section 3.2, the error-growth ratio can be simplified as $\Delta v_i^{k+1}/\Delta v_i^k \approx v_i^{k+1}/v_i^k$, where v_i^k is the magnitude of velocity. When v_i^{k+1}/v_i^k is significantly larger than 1, an instability can be identified. The detailed derivation for this ratio is presented in the Appendix.

Considering that the accumulative velocity is always in the direction of $\mathbf{b}_i$ (Fig. 10), if particle i has a velocity $\mathbf{u}_i^k$ at a time step k , the velocity at next time step will be

$$\mathbf{u}_i^{k+1} = \left(1 + \frac{l_0}{r_e} R_i\right) \mathbf{u}_i^k \text{ or } v_i^{k+1} = \left(1 + \frac{l_0}{r_e} R_i\right) v_i^k, \quad (46)$$

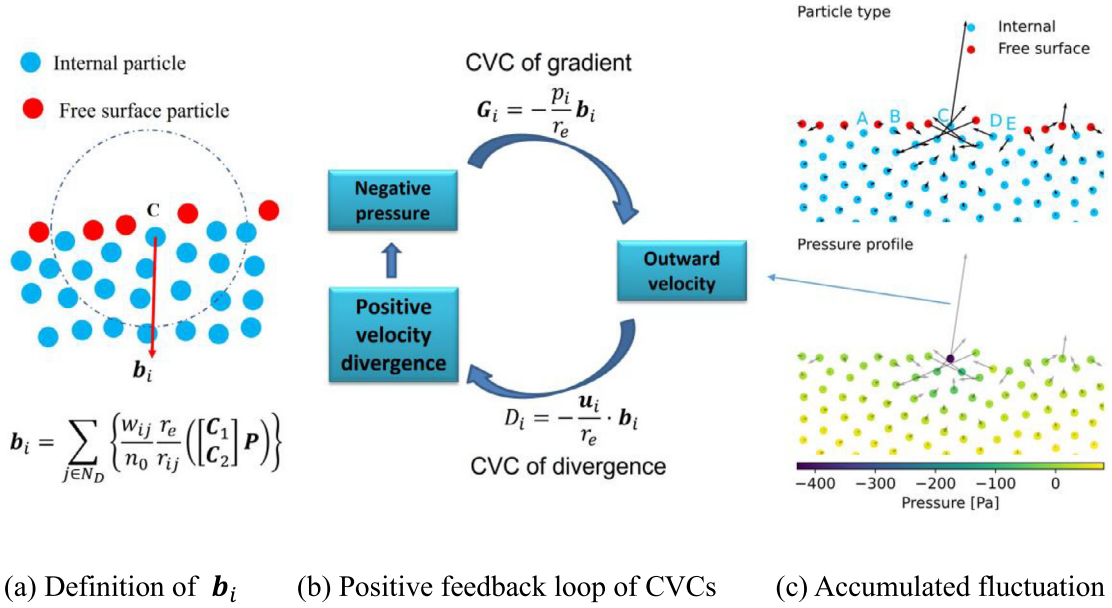


Fig. 11. The positive feedback loop of the center-variable components (CVCs) for a particle with biased neighbor support (e.g., particle C in Fig. 6).

where $v_i^k = \|\mathbf{u}_i^k\|$ and the index R_i is defined as:

$$R_i = \frac{\|\mathbf{b}_i\|^2}{c_i}. \quad (47)$$

In Eq. (46), the velocity growth depends on not Δt but R_i , clearly indicating error accumulation. For this reason, R_i is called an *error-accumulation index*. When R_i is large, CVCs always has a tendency to cause error accumulation. Namely, the recovery capability of the consistent schemes to fluctuations is rather poor on biased particle distributions. Besides, negative R_i can happen due to negative c_i . In this situation, the accumulated velocity may be frequently reversed, and the error magnitude might increase or decrease (see Section 3.4).

The proposed R_i is effective to explain why only particle C suffered from error accumulation in Fig. 6. The values of $\|\mathbf{b}_i\|$, c_i , and R_i of Points A–E are summarized in Table 2. The indices of particles A, D, and E were in the normal range without instability risk. Even though particle B took a large $\|\mathbf{b}_i\|$, R_i was still small due to the large c_i . Only particle C took a much larger R_i (around 17) than other particles (less than 0.6). This large R_i was the reason for error accumulation.

The velocity growth ratio v_i^{k+1}/v_i^k was calculated for particle C. In Table 2, the observed v_i^{k+1}/v_i^k is around 1.2, while the predicted one by Eq. (46) is around 6.5. This happens because NVCs have not been considered, as discussed below.

3.4. Influence of neighbor-variable components on error accumulation

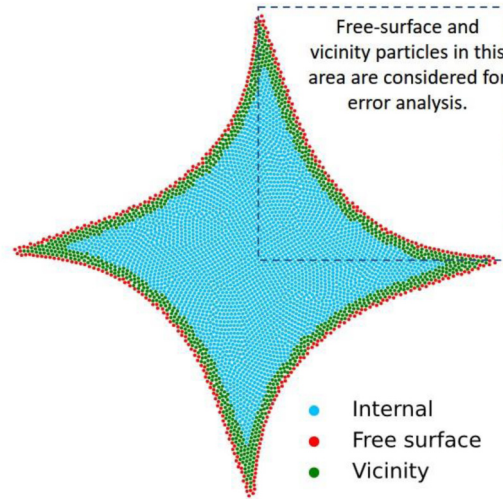
The theoretical analysis of NVCs is challenging because it is equivalent to solving governing equations analytically. Therefore, this problem is considered in a numerical and empirical manner.

The numerical experiments are based on the square-patch problem (see details in Section 5.2). The particle distribution at $t = 1$ in Fig. 12 is used. A single BEAP was produced as follows. Specifically, the free-surface and vicinity particles were selected as treatment particles one by one. In a numerical test, the treatment particle was always set as an internal particle, and the recommended free-surface-detection conditions [see Eq. (66)] were applied to the remaining particles. In this manner, only the treatment particle might be subjected to a BEAP. The velocity and indices were collected for analysis. The velocity growth ratio v_i^{k+1}/v_i^k was used to indicate instability. Specifically, the velocity change after one time step is tiny in normal situations. Thus, the ratio v_i^{k+1}/v_i^k was essentially around

Table 2

The calculated indices of $\|b_i\|$, c_i , and R_i for points A–E in Fig. 6 during the error-accumulation process. (v_i^k is the velocity magnitude, and v_i^{k+1}/v_i^k measures the velocity change after only one time step.)

Time step label		1st	2nd	3rd	4th	5th	6th	7th
Point A	$\ b_i\ $	0.735	0.732	0.728	0.725	0.722	0.719	0.715
	c_i	7.031	7.027	7.023	7.02	7.016	7.013	7.009
	R_i	<u>0.077</u>	<u>0.076</u>	<u>0.076</u>	<u>0.075</u>	<u>0.074</u>	<u>0.074</u>	<u>0.073</u>
Point B	$\ b_i\ $	2.203	2.197	2.19	2.182	2.173	2.163	2.151
	c_i	8.83	8.823	8.815	8.805	8.794	8.781	8.766
	R_i	<u>0.55</u>	<u>0.547</u>	<u>0.544</u>	<u>0.541</u>	<u>0.537</u>	<u>0.533</u>	<u>0.528</u>
Point C	$\ b_i\ $	3.56	3.554	3.555	3.564	3.584	3.617	3.666
	c_i	0.806	0.814	0.809	0.789	0.752	0.692	0.604
	R_i	15.728	15.513	15.615	16.095	17.092	18.906	22.244
	v_i^{k+1}/v_i^k	1.35	1.272	1.234	1.214	1.206	1.208	1.222
Point D	$\ b_i\ $	0.589	0.647	0.708	0.771	0.837	0.907	0.983
	c_i	5.047	4.975	4.899	4.818	4.732	4.638	4.535
	R_i	<u>0.069</u>	<u>0.084</u>	<u>0.102</u>	<u>0.123</u>	<u>0.148</u>	<u>0.178</u>	<u>0.213</u>
Point E	$\ b_i\ $	0.694	0.68	0.666	0.653	0.641	0.63	0.62
	c_i	4.937	4.956	4.973	4.989	5.003	5.015	5.025
	R_i	<u>0.097</u>	<u>0.093</u>	<u>0.089</u>	<u>0.085</u>	<u>0.082</u>	<u>0.079</u>	<u>0.077</u>

**Fig. 12.** Sketch for initial particle distribution for numerical analysis of error evolution.

1, considering the non-zero velocity near the free surface. However, when error accumulation happened, the ratio v_i^{k+1}/v_i^k would significantly deviate from 1. Like particle C in Table 2, the variables, $\|b_i\|$, c_i , R_i , and v_i^{k+1}/v_i^k , could be collected for each treatment particle suffering from BEAP. The results are portrayed in Fig. 13.

Fig. 13(a) presents the relation between v_i^{k+1}/v_i^k and R_i . The absolute value of R_i was used because the x -axis was of a logarithmic scale. When $|R_i| < 1.76$, the simulation was stable. When $|R_i| > 1.76$, the fast velocity growth could trigger instability. Therefore, the critical R_i was 1.76. The relation between v_i^{k+1}/v_i^k and $\|b_i\|$ is shown in Fig. 13(b), where the critical $\|b_i\|$ is 2.16. Namely, $\|b_i\|$ can also be regarded as an indicator for error accumulation, because the large R_i is mainly caused by $\|b_i\|$. The relation between v_i^{k+1}/v_i^k and c_i is shown in Fig. 13(c), where $c_i = 2.67$ was just a vague criterion for stability. This happened because c_i fluctuated seriously on biased particle distributions. Briefly, R_i and $\|b_i\|$ are effective indices to evaluate the error-accumulation risk.

The influence of NVCs on error accumulation is shown in Fig. 13(a), where the blue solid line is the theoretical v_i^{k+1}/v_i^k in Eq. (46). When $1.76 < R_i < 7.0$, error accumulation took place, but the observed v_i^{k+1}/v_i^k was

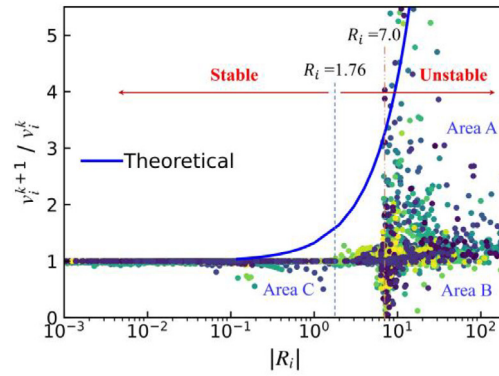
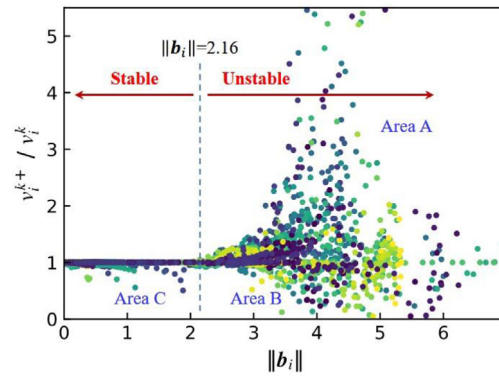
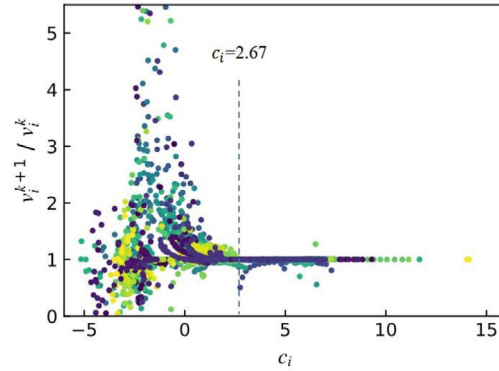
(a) v_i^{k+1}/v_i^k versus $|R_i|$ (b) v_i^{k+1}/v_i^k versus $\|b_i\|$ (c) v_i^{k+1}/v_i^k versus c_i

Fig. 13. The relation between the velocity growth ratio after one time step (v_i^{k+1}/v_i^k) and the proposed indices (R_i , $\|b_i\|$, and c_i). The points with the same color belong to the same particle. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

significantly smaller than the theoretical predication. This phenomenon clearly indicated that the NVCs alleviated error accumulation. Physically, NVCs always transferred errors to neighboring particles, thus reducing the largest error at the center. When $R_i < 1.76$, error attenuation was even observed in the Area C due to dominant NVCs. Once again, the NVCs could suppress error accumulation, and the suppression capability was strong in this situation. When $R_i > 7.0$, the NVCs usually attempted to alleviate but sometimes even accelerated error accumulation. Namely, the

alleviation capability seemed weak, even resulting in the opposite effect. Briefly, the NVCs would usually attempt to alleviate error accumulation. As the increase of R_i , the alleviation effect of NVCs became weaker and weaker.

Therefore, the error-accumulation mechanism is as follows. As the increase of R_i , the error-attenuation effect of NVCs became weaker and weaker, but the error-accumulation effect of CVCs became stronger and stronger. When the CVC effect just exceeded the NVC effect, error accumulation would occur. However, the critical value of R_i for stability was not obtained analytically. Nevertheless, the proposed index R_i that could effectively measure the error-accumulation risk was still an important progress. The critical value could be obtained by numerical tests (like Fig. 13) or the trial-and-error method.

The typical error behaviors after one time step in the Areas A, B, and C in Fig. 13 are discussed below. First, the error behaviors in Area A are presented in Fig. 14(a) and (b). When R_i was large and positive, error accumulated in the same direction in Fig. 14(a). When R_i became negative and large, the velocity error might further increase in magnitude but in the opposite direction [Fig. 14(b)]. Meanwhile, the velocity error might decrease after the direction was reversed, when R_i was negative, as presented in Fig. 14(c). This corresponded to Area B in Fig. 13. When R_i was further reduced in Fig. 14(d), error attenuation was observed, due to dominant NVCs. This corresponded to Area C of Fig. 13. These error behaviors agree well with the theoretical analysis in Section 3.3.

Next, the intermittent BEAP interactions are discussed below to mimic the real error behaviors in dynamic simulations. The reason for intermittence is that free surface particles are wrongly detected only on rare occasions. The reason for interaction is that the large errors due to a BEAP can easily result in misdetection of nearby particles. During intermittent interactions, error transfer would play a significant role. The numerical tests were still based on the case in Fig. 12. Two BEAPs were produced by setting two adjacent free-surface treatment particles as internal particles simultaneously. This special treatment was turned off after significant error accumulation to mimic intermittence. Several examples of the BEAP interactions are shown in Fig. 15, where the particles A and B were the treatment ones. Obviously, the velocity fluctuations were rather random, and they were quite different from the BEAP ones in Fig. 14. The strong and continuous error transfer during many time steps were the key reasons for these phenomena. Similar spurious currents due to intermittent BEAP interactions were observed in Fig. 1(a, b). Particularly, the reproduced fluctuations in Fig. 15(b) were essentially like Fig. 1(b). Briefly, when intermittent BEAP interactions happened, there was basically no definite pattern for fluctuations. The only remaining feature was the local and large fluctuations.

Finally, the proposed viewpoint on error accumulation is used to analyze the stability strategies near free surfaces in conventional MPS. The above analysis can also be applied to the original MPS models. For the standard gradient model:

$$\langle \nabla p \rangle_i = \frac{d}{n_0} \sum_{j \neq i} \left\{ w_{ij} \frac{p_j - p_i}{r_{ij}} \frac{\mathbf{r}_{ij}}{r_{ij}} \right\}, \quad (48)$$

where d is the dimension number, the CVC that can cause error accumulation is:

$$\mathbf{G}_i = -p_i \frac{d}{n_0} \sum_{j \neq i} \left\{ w_{ij} \frac{1}{r_{ij}} \frac{\mathbf{r}_{ij}}{r_{ij}} \right\}. \quad (49)$$

For the particles near free surfaces, the stable gradient models are

$$\langle \nabla p \rangle_i = \frac{d}{n_0} \sum_{j \neq i} \left\{ w_{ij} \frac{p_j}{r_{ij}} \frac{\mathbf{r}_{ij}}{r_{ij}} \right\} = \frac{d}{n_0} \sum_{j \neq i} \left\{ w_{ij} \frac{p_j - p_i}{r_{ij}} \frac{\mathbf{r}_{ij}}{r_{ij}} \right\} + p_i \frac{d}{n_0} \sum_{j \neq i} \left\{ w_{ij} \frac{1}{r_{ij}} \frac{\mathbf{r}_{ij}}{r_{ij}} \right\}, \quad (50)$$

and

$$\langle \nabla p \rangle_i = \frac{d}{n_0} \sum_{j \neq i} \left\{ w_{ij} \frac{p_j + p_i}{r_{ij}} \frac{\mathbf{r}_{ij}}{r_{ij}} \right\} = \frac{d}{n_0} \sum_{j \neq i} \left\{ w_{ij} \frac{p_j - p_i}{r_{ij}} \frac{\mathbf{r}_{ij}}{r_{ij}} \right\} + 2p_i \frac{d}{n_0} \sum_{j \neq i} \left\{ w_{ij} \frac{1}{r_{ij}} \frac{\mathbf{r}_{ij}}{r_{ij}} \right\}. \quad (51)$$

As pointed out by Shibata et al. [33], the first model is actually the one using minimal pressure in neighborhood p_{min} [49], because of $p_{min} = 0$ near free surfaces. The second model is the conservative one [50]. Compared to Eq. (48), the modification terms are added in Eqs. (50) and (51). These modification terms just help to cancel out the CVC that can easily cause error accumulations in Eq. (49). Thus, the present error-accumulation mechanism is consistent with conventional stabilization terms. Note that the modification terms are not employed in the present gradient model.

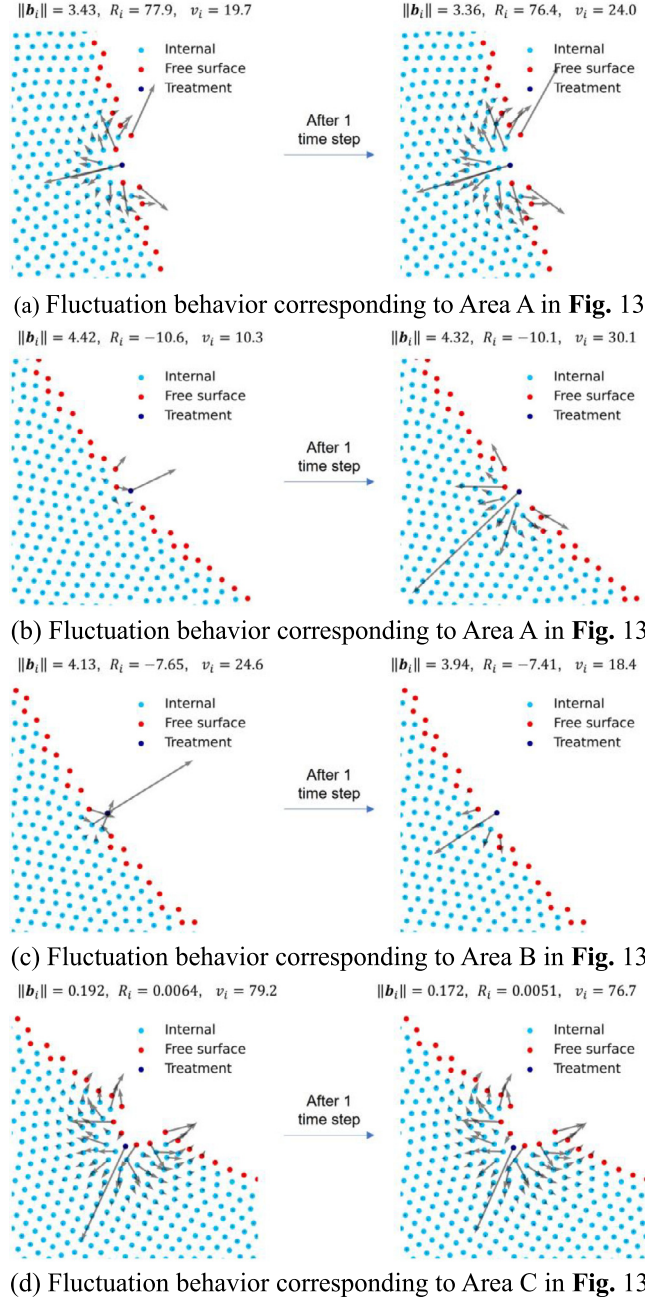


Fig. 14. Typical changes of velocity errors after one time step corresponding to the Areas A, B, and C in Fig. 13.

3.5. Error evolution analysis for the diffusion process and PPE convergence

After the consistent schemes were employed, we encountered the divergence of solving PPE on rare occasions. This happened probably because unphysical negative coefficients (i.e., negative $[C_3 + C_4]P$) took place in the Laplacian model on the biased neighbor support. The iterative process of solving PPE is like the dynamic diffusion to a steady state. Therefore, the influence of the negative coefficients on error evolution of a diffusion process is considered here to give hints for the PPE convergence.

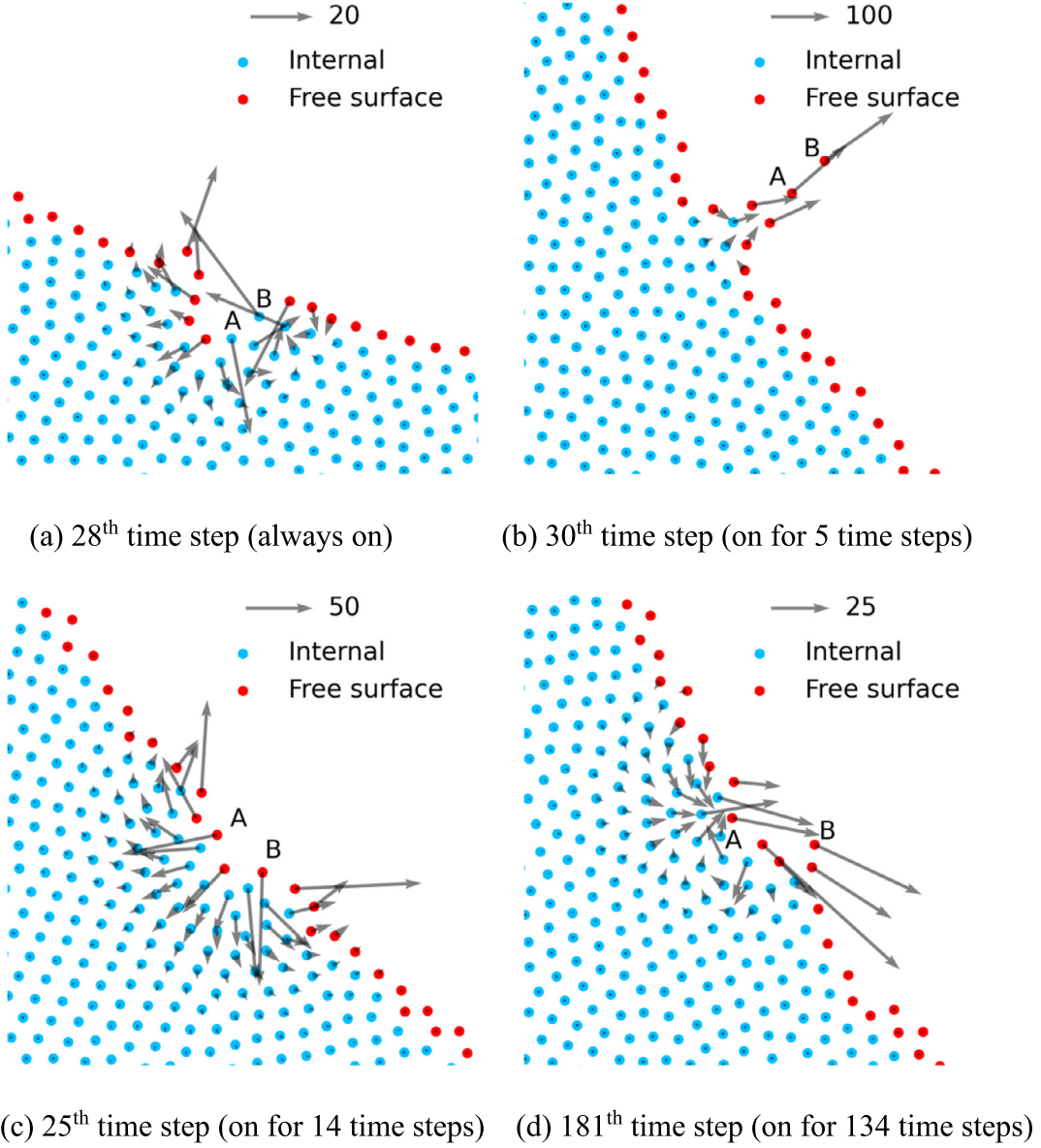


Fig. 15. Velocity fluctuation examples produced by the intermittent interaction of the two BEAPs. The information in the parentheses indicates how long the treatment lasted. For instance, “on for 5 time steps” indicates the treatment is kept on for 5 time steps.

Solving PPE is to find the steady solution of a diffusion problem:

$$\frac{\partial p}{\partial \tau} = \frac{1}{\rho} \nabla^2 p + S, \quad (52)$$

where τ is the artificial time to consider variable dimensions, and S is the given PPE source. From the Laplacian model [Eq. (13)], the diffusion equation is discretized into

$$\frac{p_i^{k+1} - p_i^k}{\Delta \tau} = \frac{2}{\rho_i n_0} \sum_{j \in N_D} \left\{ w_{ij} \frac{p_j^k - p_i^k}{r_{ij}} \frac{[C_3 + C_4] \mathbf{P}}{l_0} \right\} + S_i. \quad (53)$$

Let Δp_i and Δp_j be the pressure errors of particles i and j , respectively. The simulated pressure can then be split into:

$$\begin{cases} p_i = \bar{p}_i + \Delta p_i \\ p_j = \bar{p}_j + \Delta p_j \end{cases}, \quad (54)$$

where $\bar{p}_i$ and $\bar{p}_j$ are the theoretical pressures. Inserting Eq. (54) into Eq. (53), one can obtain:

$$\frac{\bar{p}_i^{k+1} - \bar{p}_i^k + \Delta p_i^{k+1} - \Delta p_i^k}{\Delta \tau} = \frac{2}{\rho_i n_0} \sum_{j \in N_D} \left\{ w_{ij} \frac{\bar{p}_j^k - \bar{p}_i^k + \Delta p_j^k - \Delta p_i^k}{r_{ij}} \frac{[C_3 + C_4] \mathbf{P}}{l_0} \right\} + S_i. \quad (55)$$

Because the theoretical pressure and the source term satisfy the governing equation (52), they automatically disappear. The remaining parts become the error transfer equation:

$$\frac{\Delta p_i^{k+1} - \Delta p_i^k}{\Delta \tau} = \frac{2}{\rho_i n_0} \sum_{j \in N_D} \left\{ w_{ij} \frac{\Delta p_j^k - \Delta p_i^k}{r_{ij}} \frac{[C_3 + C_4] \mathbf{P}}{l_0} \right\}. \quad (56)$$

Let Γ_{ij} denote the Laplacian-model coefficient between a pair of particles as follows

$$\Gamma_{ij} = \frac{2w_{ij} [C_3 + C_4] \mathbf{P}}{n_0 r_{ij} l_0}. \quad (57)$$

From this definition, Eq. (56) can be simplified into

$$\frac{\Delta p_i^{k+1} - \Delta p_i^k}{\Delta \tau} = \frac{1}{\rho_i} \sum_{j \in N_D} \{ (\Delta p_j^k - \Delta p_i^k) \Gamma_{ij} \}. \quad (58)$$

Then, the error of particle i at next time Δp_i^{k+1} can be obtained as

$$\Delta p_i^{k+1} = \Delta p_i^k - \frac{\Delta \tau}{\rho_i} \Delta p_i^k \sum_{j \in N_D} \{ \Gamma_{ij} \} + \frac{\Delta \tau}{\rho_i} \sum_{j \in N_D} \{ \Delta p_j^k \Gamma_{ij} \}. \quad (59)$$

Next, the error-growth ratio after one time step can be calculated from

$$\frac{\Delta p_i^{k+1}}{\Delta p_i^k} = 1 - \frac{\Delta \tau}{\rho_i} \sum_{j \in N_D} \{ \Gamma_{ij} \} + \frac{\Delta \tau}{\rho_i} \sum_{j \in N_D} \left\{ \left(\frac{\Delta p_j^k}{\Delta p_i^k} \right) \Gamma_{ij} \right\}. \quad (60)$$

Up to now, Eq. (60) holds exactly. Generally, the error terms, Δp_i^k and Δp_j^k , may be of the same order but probably with different signs. In this situation, one can assume

$$\frac{\Delta p_j^k}{\Delta p_i^k} \approx \pm 1. \quad (61)$$

If the particle distributions are basically isotropic, Γ_{ij} will always be positive, corresponding to the physical positive diffusion. In this situation, the second part of right-hand-side (RHS) of Eq. (60) is a dissipation term that always reduces $\Delta p_i^{k+1}/\Delta p_i^k$. The third part of RHS of Eq. (60) is a fluctuation term that always transfers neighboring errors to the center particle. When $\Gamma_{ij} > 0$, based on Eqs. (60) and (61), one has

$$\frac{\Delta \tau}{\rho_i} \sum_{j \in N_D} \{ \Gamma_{ij} \} > \frac{\Delta \tau}{\rho_i} \sum_{j \in N_D} \{ \pm \Gamma_{ij} \} \text{ and thus } \frac{\Delta p_i^{k+1}}{\Delta p_i^k} < 1, \quad (62)$$

which indicates that the error always decreases with time. In this case, the PPE will be well conditioned and stably convergent. However, when the particle distribution is seriously biased, negative Γ_{ij} that corresponds to unphysical negative diffusion will take place. If $\sum_{j \in N_D} \{ \Gamma_{ij} \} < 0$, one cannot guarantee that $\Delta p_i^{k+1}/\Delta p_i^k$ is generally smaller than 1. Meanwhile, the PPE system may not be well conditioned, and its convergence becomes difficult. In brief, negative Γ_{ij} may increase the diffusion error over time and make the PPE system difficult to converge.

Next, two indices are proposed to measure the influence of negative Γ_{ij} . The first one is the index c_i ,

$$c_i = \sum_{j \in N_D} \{ (r_e l_0) \Gamma_{ij} \}. \quad (63)$$

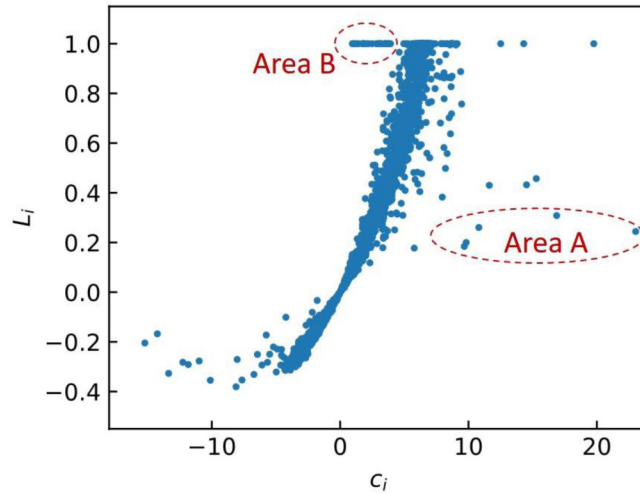


Fig. 16. Relation between c_i and L_i for the free surface particles in the square-patch problem.

It measures how much the dissipation term is reduced by negative Γ_{ij} . The maximum fluctuation caused by the fluctuation term in Eq. (60) is proportional to

$$\llbracket c \rrbracket_i = 2 \sum_{j \in N_D} \{ |(r_e l_0) \Gamma_{ij}| \}. \quad (64)$$

Then, a Laplacian ratio can be defined from c_i and $\llbracket c \rrbracket_i$ as follows:

$$L_i = \frac{c_i}{\llbracket c \rrbracket_i} = \frac{\sum_{j \in N_D} \{ \Gamma_{ij} \}}{\sum_{j \in N_D} \{ |\Gamma_{ij}| \}}. \quad (65)$$

This ratio measures the relative scales of the diffusion and fluctuation terms. L_i is 1 for normal internal particles but decreases rapidly near free surface. The difference between c_i and L_i can be found in Fig. 16. Specifically, the circled particles in Area A have large c_i but take small L_i . These situations happen when the positive and negative coefficients Γ_{ij} fluctuates seriously. On the other hand, the circled particles in Area B exhibit normal $L_i = 1$ but take small c_i . These situations happen when the special modification in Section 2.2 is applied. Briefly, c_i and L_i are not interchangeable.

Lastly, numerical examples were performed to show the influence of negative Γ_{ij} on the PPE convergence. Two groups of tests were conducted as follows. The particle distribution at $t = 1$ in Fig. 12 was employed. The PPE was solved for one time step, and the required iteration number for converged PPE was collected. In the control group, all the free surface particles were normally detected from the recommended conditions [Eq. (66)]. Therefore, all the particles in the PPE system took positive c_i (and positive L_i). In the treatment group, the free surface particles with negative c_i (or negative L_i) in Fig. 12 were selected as the internal treatment particles one by one. In this manner, the influence of one particle with negative c_i on the PPE convergence is investigated. The results at $t = 1$ are contrasted in Fig. 17(a), where the treatment group suffered from serious fluctuations in the required iteration number. Then, such tests were performed at $t = 0.5, 1.5, 2.0$ and 2.5 . The comparison of the mean convergent number is shown in Fig. 17(b), where the treatment group required a higher mean iteration number than the control group.

In brief, the inclusion of one particle with unphysical negative diffusion into the PPE system could result in fluctuations and increase the required iteration number, indicating that PPE became more difficult to converge. Therefore, it is necessary to restrict c_i and L_i in free-surface detection to (1) suppress unphysical negative diffusion and (2) guarantee the well-conditioned PPE system.

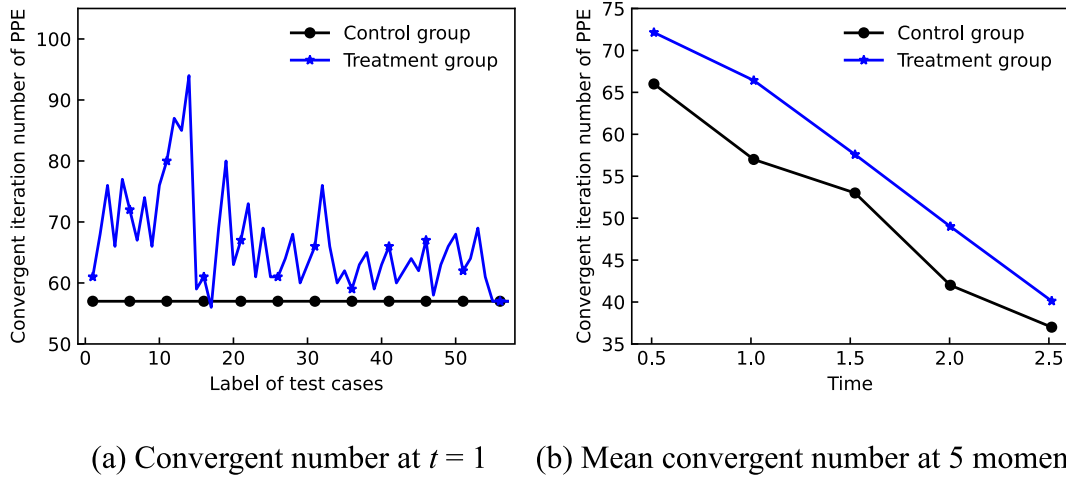


Fig. 17. Comparison of convergent iterative number of PPE between the control group (no negative c_i) and the treatment group (containing one particle with negative c_i).

4. Novel conditions for free surface detection based on error analysis

The influence of free surface detection on error accumulation is discussed below. As shown in Fig. 11, the error accumulation depends on a positive feedback loop of CVCs containing u_i and p_i . When a particle is detected as a free surface one, p_i becomes a constant value without fluctuations, breaking the positive feedback loop. Therefore, error accumulation stops, even though the neighbor support is still incomplete. Namely, if the particles with biased neighbor supports are promptly detected as free-surface ones, fast error accumulation will not happen. Otherwise, error accumulation will trigger unphysical currents. Therefore, the paper provides a clear criterion for free-surface detection: the particles with high error-accumulation risk must be detected as free-surface ones to guarantee the stability.

Even though the thresholds were not obtained theoretically, the proposed indices are still effective to provide stable regions without error accumulation (see Fig. 13). Thus, the indices can be employed to enhance the stability via the free-surface detection. Briefly, the recommended conditions for free surface detection are:

$$\begin{cases} \text{if } (\|b_i\| > 2.0 \text{ or } R_i > 1.5 \text{ or } c_i < 3.0 \text{ or } L_i < 0.25) & \text{Free surface particle} \\ \text{else if } (N_i < 0.35N_0) & \text{Parachute region detection} \\ \text{else} & \text{Internal particle} \end{cases}, \quad (66)$$

where “parachute-region-detection” was defined in Section 2.3. Even though only two of $\|b_i\|$, c_i , and R_i are independent, restrictions are provided for all three indices to find the thresholds more easily by trial and error. A sufficient free-surface-detection condition must consider all kinds of instabilities. The reasons for each condition are as follows. The restriction on R_i is used to suppress the fast error accumulation. The restriction on $\|b_i\|$ is used to suppress the large bias error due to sudden pressure fluctuations when free surfaces impact each other. The restrictions on c_i and L_i are used to suppress unphysical negative diffusion and guarantee a well-conditioned PPE system (Section 3.5). The “parachute-region-detection” is used to suppress the fluctuations at free surfaces due to the reduced discretization accuracy (see Fig. 4(b) or Section 2.2) when the neighboring-particle number significantly decreases. In this situation, guaranteeing that the internal particles are sufficiently covered by free-surface ones can help to suppress the unphysical large pressure-gradient forces. The thresholds for R_i and $\|b_i\|$ can be determined from Fig. 13, and those for c_i , L_i , and “parachute-region detection” are determined empirically.

Lastly, there are two notes for the proposed conditions. First, via comparing Eqs. (66) and (16), all the free-surface particles detected by the second line of Eq. (66) will be successfully detected by the existing conditions [Eq. (16)]. Namely, the performance difference between the proposed and existing conditions in Section 5 is only caused by the proposed indices. Second, the detection conditions in Eq. (66) suppress not only a single BEPA but also the interactions among BEAPs.

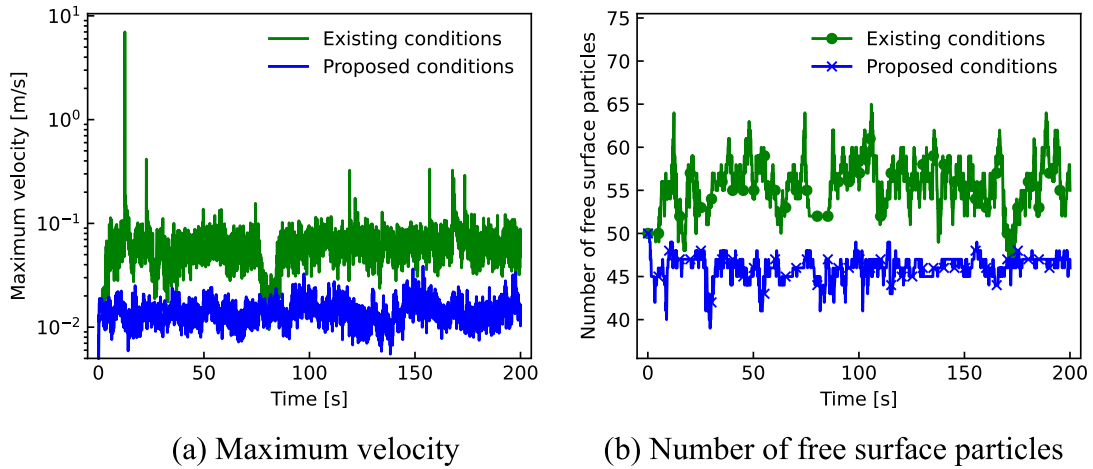


Fig. 18. Comparison of maximum velocity (left) and free-surface detection (right) simulated by the existing and proposed conditions in the hydrostatic-pressure problem.

5. Results and discussion

In this section, four free-surface flow cases were simulated: a hydrostatic-pressure problem, a square-patch problem, a dam break flow, and an inflow impact problem. The performance of the proposed conditions for the positive/negative pressure under free surfaces was discussed in the first three cases. The capability for the topological changes of free surfaces was investigated in the last two cases. To contrast the stability performance between the existing (Section 2.3) and proposed (Section 4) conditions, all other parameters were kept the same. The interaction radius $r_e = 3.1 \times l_0$ was adopted.

5.1. Hydrostatic-pressure problem

The first case is a hydrostatic-pressure problem where positive pressure occurs under the free surface. It is relatively easy to keep the simulation stable. The following problem is considered. A water patch of size $0.1 \text{ m} \times 0.1 \text{ m}$ is simulated. The density is 1000 kg/m^3 , the viscosity is 0.001 Pa s , and the gravity is 10 m/s^2 . The flow is simulated for a long time (200 s) with an average interparticle distance $l_0 = 0.002 \text{ m}$. Slight random fluctuations (up to $0.01l_0$) are added to the initial particle positions to trigger the velocity fluctuations.

First, the maximum-velocity histories simulated by the existing and proposed conditions are contrasted. As shown in Fig. 18(a), the maximum velocity using the existing conditions was much larger than that using the proposed conditions. Particularly, instability happened at $t = 12.5 \text{ s}$ when the existing conditions were adopted, as shown in Fig. 19(a). This unstable area is enlarged and presented in Fig. 1(a). After a while, the simulation became stable again [Fig. 18(a)], but severe free-surface fluctuations occurred easily [Fig. 19(b)]. By contrary, the proposed conditions produced much smaller fluctuations in terms of the velocity and the free surface position than the existing ones, as shown in Fig. 19(c) and (d).

The proposed conditions effectively suppressed error accumulation compared to the existing ones. Thus, the observed fluctuation difference was attributed to error accumulation. Specifically, some free-surface misdetections could occasionally take place for the existing conditions. If the misdetections lasted for a relatively long time (e.g., several time steps), severe error accumulation would happen, causing large and local fluctuations [Fig. 19(a)]. In this process, the severe fluctuations could cause the misdetection of more particles, resulting in BEAP interactions (See Fig. 15). If the misdetections just lasted for a relatively short time (e.g., one or two time steps), minor error accumulation would take place, just causing additional fluctuations. Then, the error transfer would gradually diffuse the fluctuations everywhere near the free surface [Fig. 19(b)]. Briefly, the observed fluctuations due to error accumulation are usually influenced by the BEAP interactions or sufficiently diffused by error transfer. Similar error accumulation phenomena will be observed in the following sections.

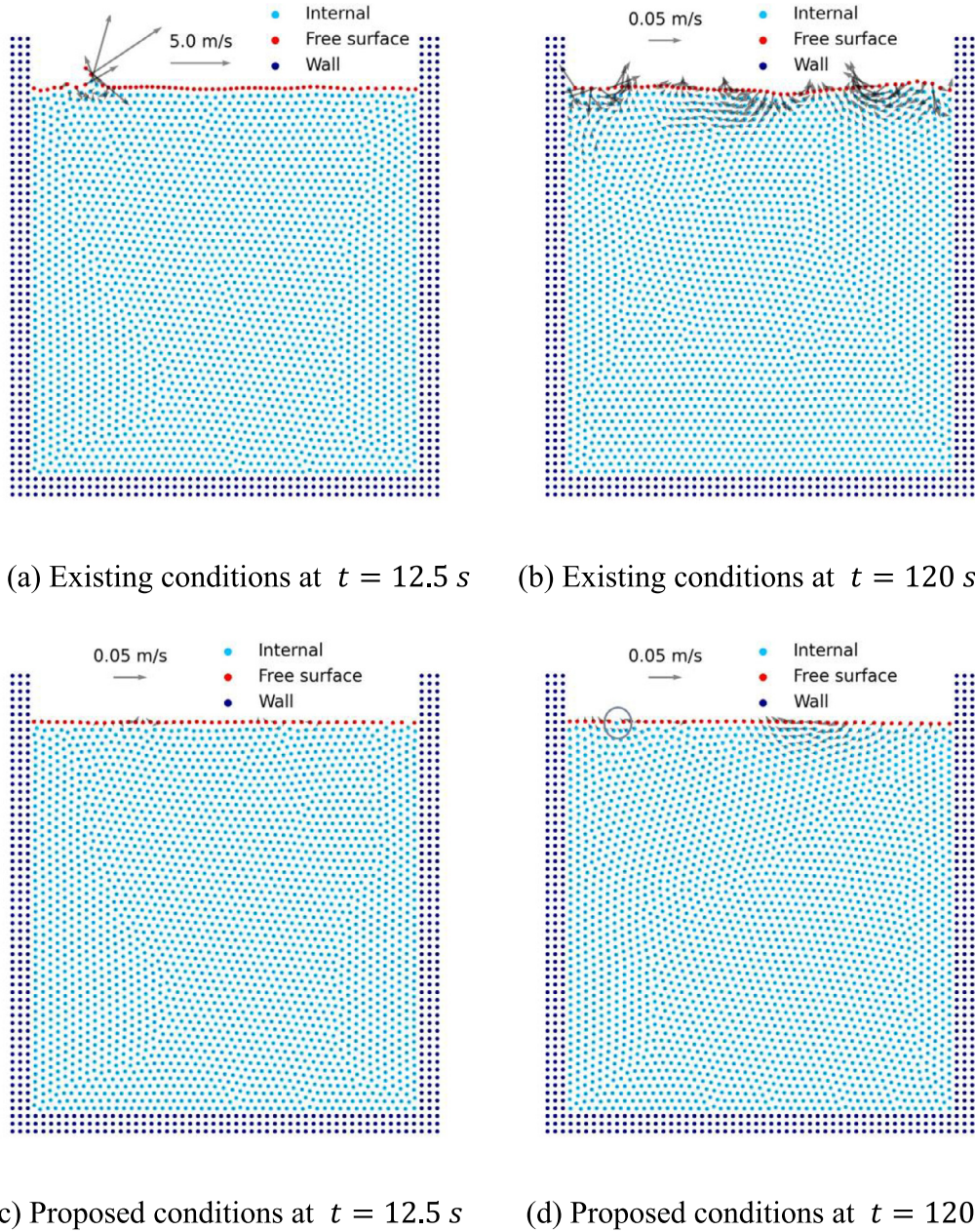


Fig. 19. Comparison of velocity distributions simulated using the existing and proposed conditions in the hydrostatic-pressure problem at two moments.

Second, the free surface detection using the existing and proposed conditions is compared. As shown in Fig. 18(b), the same number of free surface particles, namely 50, was detected initially by both conditions. After some time, the existing conditions detected more free surface particles (>50), while the proposed conditions detected less free surface particles (<50), as shown in Figs. 18(b) and 19.

In brief, an interesting phenomenon was observed: the proposed conditions detected less free surface particles but produced better stability than the existing conditions. This phenomenon was consistent with the novel idea for free surface detection in this paper. Namely, only the particles that were prone to error accumulation would be detected as free surface particles. A free-surface-detection example by the proposed conditions was shown by the circled

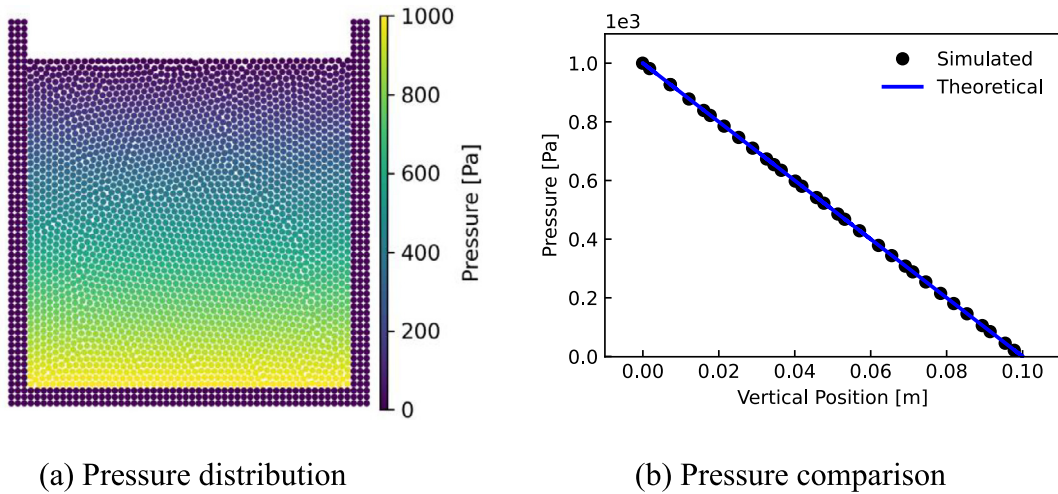


Fig. 20. Simulated pressure profile and pressure comparisons along the vertical central line at $t = 200$ s.

particle in Fig. 19(d). Even though this particle was close to free surface, it was not prone to error accumulation based on the proposed indices, and thus it would not be detected as a free surface particle. Then, it would gradually move into the internal area, and this was the reason why less free surface particles were detected. Even though the existing conditions detected more free surface particles, some particles prone to error accumulation were still not detected completely or promptly. Thus, instability could still happen occasionally, and the fluctuation level was high.

Due to the adoption of the consistent schemes, smooth pressure profile was produced, and the simulated pressure agreed well with the theoretical result, as shown in Fig. 20. Therefore, the method implementation is verified by this problem.

5.2. Square-patch problem

The second case is the square-patch problem [51], which is subject to large free-surface deformation under negative pressure. Because this is a purely numerical case, units are neglected, and SI units can be assumed. For a square patch with edge length $L_0 = 1$, the following initial velocity is specified:

$$\mathbf{u}_0(x, y) = \begin{bmatrix} \omega y \\ -\omega x \end{bmatrix}, \quad (67)$$

where $\omega = 1$. The fluid density is $\rho = 1000$, and the viscosity is neglected. Gravity is not considered. A 101×101 particle arrangement is adopted for simulation.

First, we discuss the performance of the existing and proposed conditions. The simulated histories of the maximum velocity are shown in Fig. 21(a). When the existing conditions were adopted, the maximum velocity suddenly soared around $t\omega = 2.2$, and the simulation became unstable, as also shown in Fig. 22(a). The large and local fluctuations clearly indicated fast error accumulation. The large fluctuations easily caused misdetection of the more nearby particles, causing the intermittent BEAP interactions (Fig. 15). Thus, the fluctuation distributions were random and different from those of a single BEAP in Fig. 14. Finally, the velocity increased to a rather high value, and then the simulation was terminated due to the tiny time step. On the other hand, the proposed conditions could produce stable results for a longer time than the existing ones. Once again, the proposed conditions still detected less free-surface particles but provided better stability compared to the existing ones, as shown in Fig. 21(b). This was consistent with the observation in Section 5.1.

Second, the results simulated using the proposed conditions are presented for verification. The simulated pressure distributions at two moments are presented in Fig. 23, where the pressure profiles were basically smooth. After $t\omega = 2.5$, some fluctuations with sharp angles took place at the free surface and caused some minor pressure

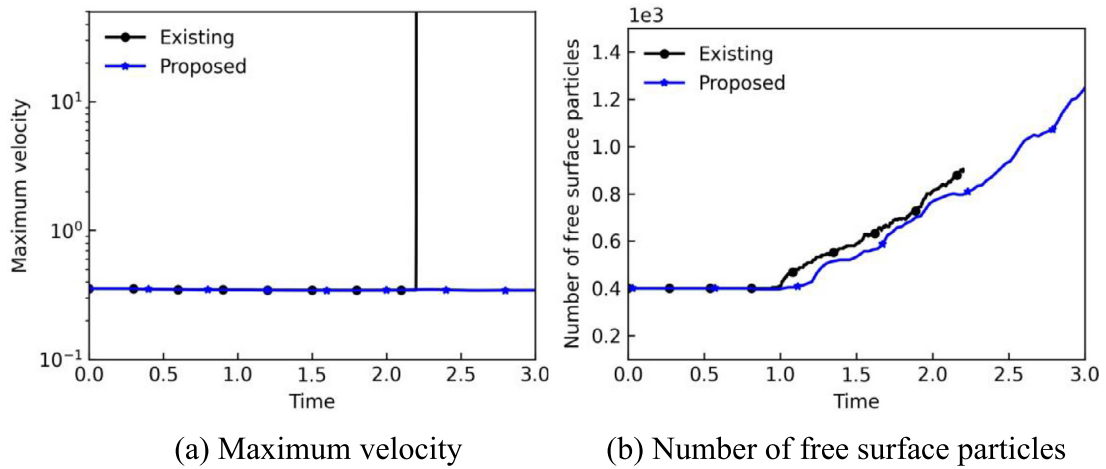


Fig. 21. Comparison of maximum velocity and free surface detection simulated by the existing and proposed conditions in the square-patch problem. Instability happened at $t\omega = 2.2$ when the existing conditions were applied.

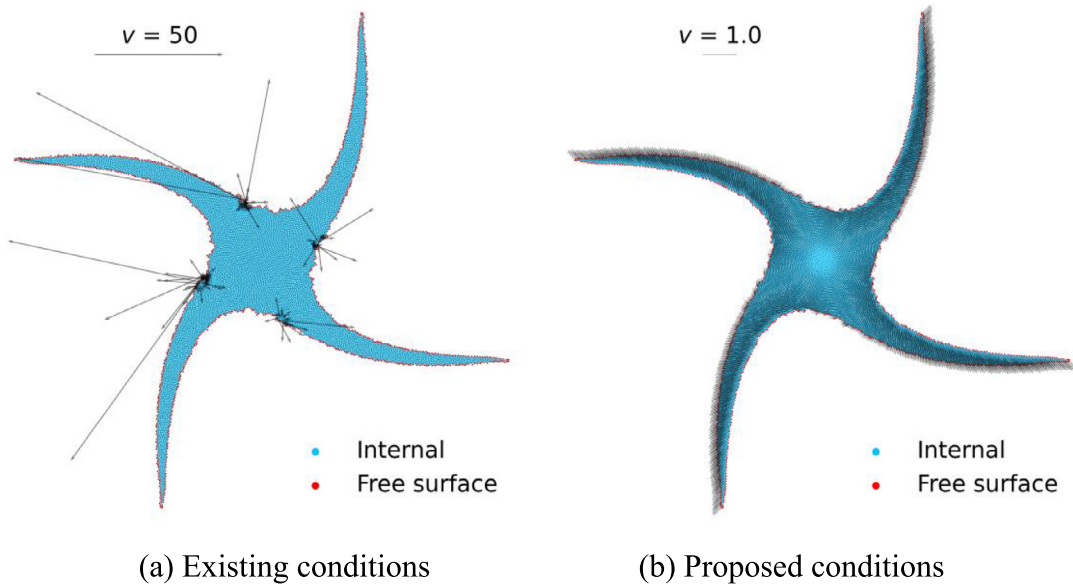


Fig. 22. Comparison of simulated velocity distributions using different conditions at $t\omega = 2.2$ in the square-patch problem.

fluctuations. As shown in Fig. 21(a), the sharp increase of velocity did not happen, and the simulation by the proposed conditions was still stable. This paper only concentrates on the stability problem due to fast error accumulation. Thus, the alleviation of this kind of free-surface fluctuations is out of the scope of this study, and it would be considered in the future works. As shown in Fig. 24, the dimensionless central pressures $p/(\rho\omega^2 L_0^2)$ agreed satisfactorily with the BEM result [51], which verified the method. The minor fluctuations after $t\omega = 2.5$ were caused by the free surface fluctuations as discussed above and shown in Fig. 23(b).

5.3. Dam break flow

The next test case is a dam break flow, which involves topological changes of free surfaces. The experiment presented in Ref. [52] is simulated here in 2D. The initial flow configuration and the measurement points are sketched in Fig. 25. Specifically, the initial water column is placed in the left side of a tank. The histories of the

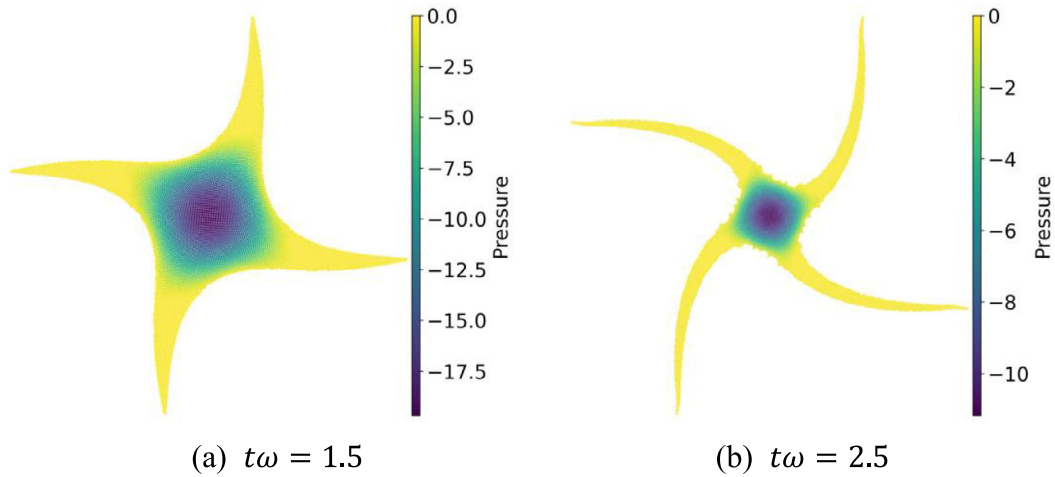


Fig. 23. Simulated pressure distributions at two different moments using the proposed conditions in the square-patch problem.

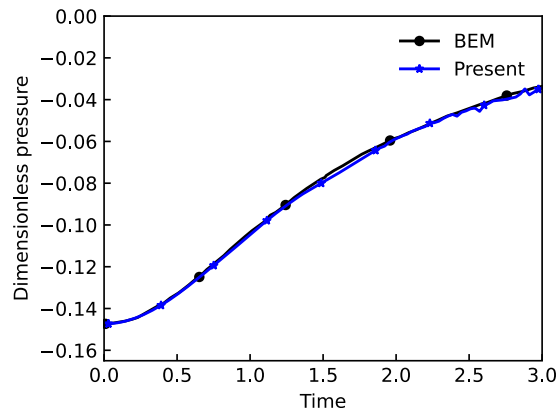


Fig. 24. Comparison of the dimensionless central pressure $[p / (\rho\omega^2 L_0^2)]$ in the square-patch problem. The proposed conditions for free surface detection were used.

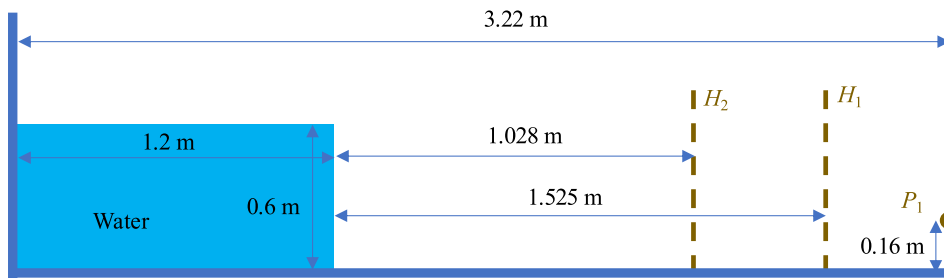


Fig. 25. Sketch of the initial configuration and measurement points in the dam break flow.

pressure at point P_1 and the water depths along the two vertical lines H_1 and H_2 were measured in the experiment. The density is $\rho = 1000 \text{ kg/m}^3$, the dynamic viscosity is $\mu = 0.001 \text{ Pa s}$, the gravity is 9.8 m/s^2 , the adopted particle size is 0.01 m , and the largest allowable time step is 0.002 s . The values for the reference pressure at free surfaces, $p_{ref} = -10^5, 0$, and 10^5 Pa , are considered.

First, we investigated the stability performance of the existing and proposed conditions. The reference pressure $p_{ref} = 0$ was adopted. The maximum-velocity histories simulated using the two kinds of conditions are shown

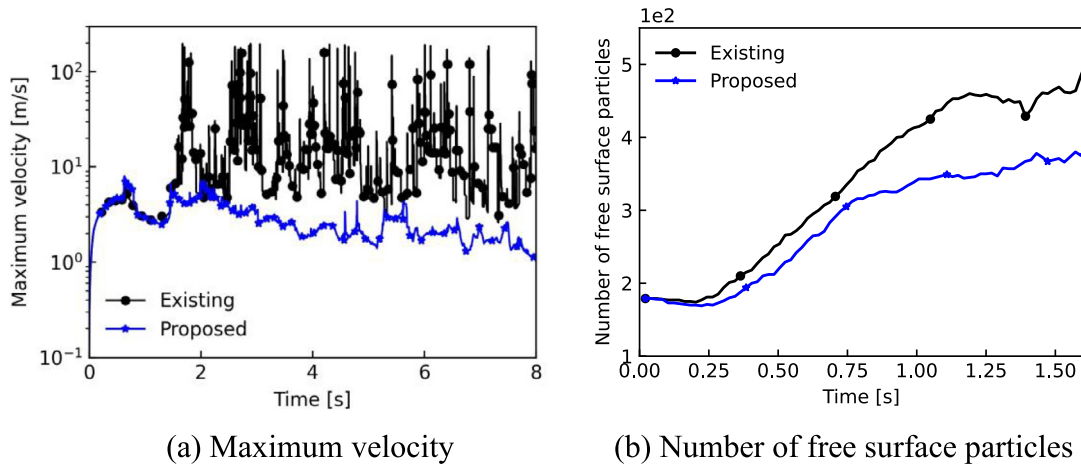


Fig. 26. Comparison of maximum velocity and free surface detection simulated by the existing and proposed conditions in the dam break flow.

in Fig. 26(a). To make the simulation continue, the largest velocity was restricted to 200 m/s. This strategy is also adopted in Section 5.4. This strategy was never invoked when the proposed conditions were applied. Both conditions produced stable results before $t = 1.5$ s. The simulated velocity distributions were also rather similar, as shown in Fig. 27(a) and (b). However, it must be noted that the proposed conditions detected much less free-surface particles than the existing ones, as shown in Fig. 26(b). After $t = 1.5$ s, when the existing conditions were applied, instability happened frequently [Fig. 26(a)]. Unphysical large velocity took place suddenly and frequently, as shown in Fig. 27(c), mainly due to the intermittent error accumulation and the continuous error transfer. On the other hand, when the proposed conditions were applied, the maximum velocity was in a reasonable range less than 10 m/s. The velocity distribution was smooth, as shown in Fig. 27(d). In brief, the proposed conditions detected less free-surface particles but produced more stable results than the existing ones.

Second, the simulated velocity distributions during dam break were presented. The proposed conditions and $p_{ref} = 0$ were adopted. The snapshots at several different moments were shown in Fig. 28, where the distributions were smooth without unreasonable large velocity. In the late phase ($t > 3.0$ s), many vortices were observed [Fig. 28(d) and (3)], probably indicating the turbulence effect was important for the energy dissipation.

Third, the influence of p_{ref} and the method validation were discussed. One feature of the present method is that only the relative pressure is involved. Hence, the simulated results are expected independent of p_{ref} . The pressure distributions simulated using different p_{ref} were compared in Fig. 29. Overall, the results of different p_{ref} were quite like each other. In the late phase around 2.13 s, slight fluctuations of the pressure and free surfaces took place, due to the splash flow instability. Particularly, the low pressure in Fig. 29(a, d) did not trigger tensile instability, and the high pressure in Fig. 29(c, f) did not seem to produce significant numerical diffusion. All the three simulations were stable. Then, the quantitative comparisons between the simulated and experimental results are presented in Fig. 30. When the pressure at P_1 was compared, p_{ref} was extracted. Considering the difference between the 2D simulation and the 3D experiment, the quantitative results simulated using different p_{ref} were in reasonable agreement with the experimental results, validating the developed method. One reason for the fluctuations after 1.5 s in Fig. 30(b–d) was the flow instability after the dam-break front impacted the right wall. Briefly, the present gradient model only depends on the relative pressure, and thus arbitrary reference pressure can be used at free surfaces.

5.4. An inflow impact problem

The last case is an inflow impact problem, where the topological changes of free surfaces become more violent than dam break. Thus, this problem was more challenging for the consistent schemes. The initial flow configuration is sketched in Fig. 31. The water is selected as the inflow fluid. Specifically, the density is $\rho = 1000$ kg/m³, the dynamic viscosity is $\mu = 0.001$ Pa s, the gravity is 9.8 m/s², the adopted particle size is 0.002 m, and the

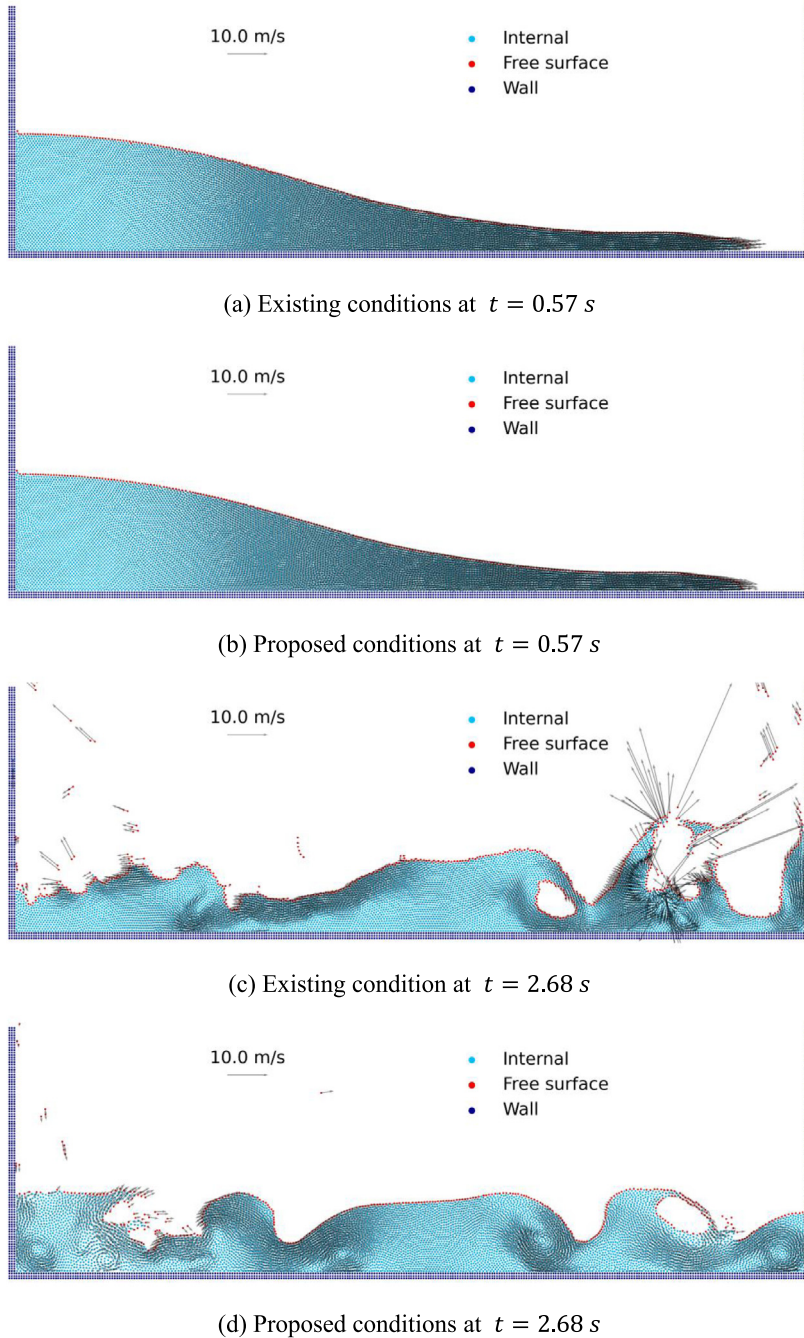


Fig. 27. Comparison of simulated velocity distribution using the existing and proposed conditions in dam-breaking simulations.

largest allowable time step is 0.001 s. Considering that (1) the viscosity is low and (2) vortices take place easily, a turbulence model similar to Ref. [53] is considered to accelerate the vortex dissipation. The particles move out of the right boundary will be deleted.

First, the stability performance is contrasted between the existing and proposed conditions. As shown in Fig. 32, the simulation using the existing conditions suffered from quite large velocity, and such unreasonable large velocity obviously reduced the simulation reliability, as shown in Fig. 33(a) and (c). Therefore, even though the turbulence

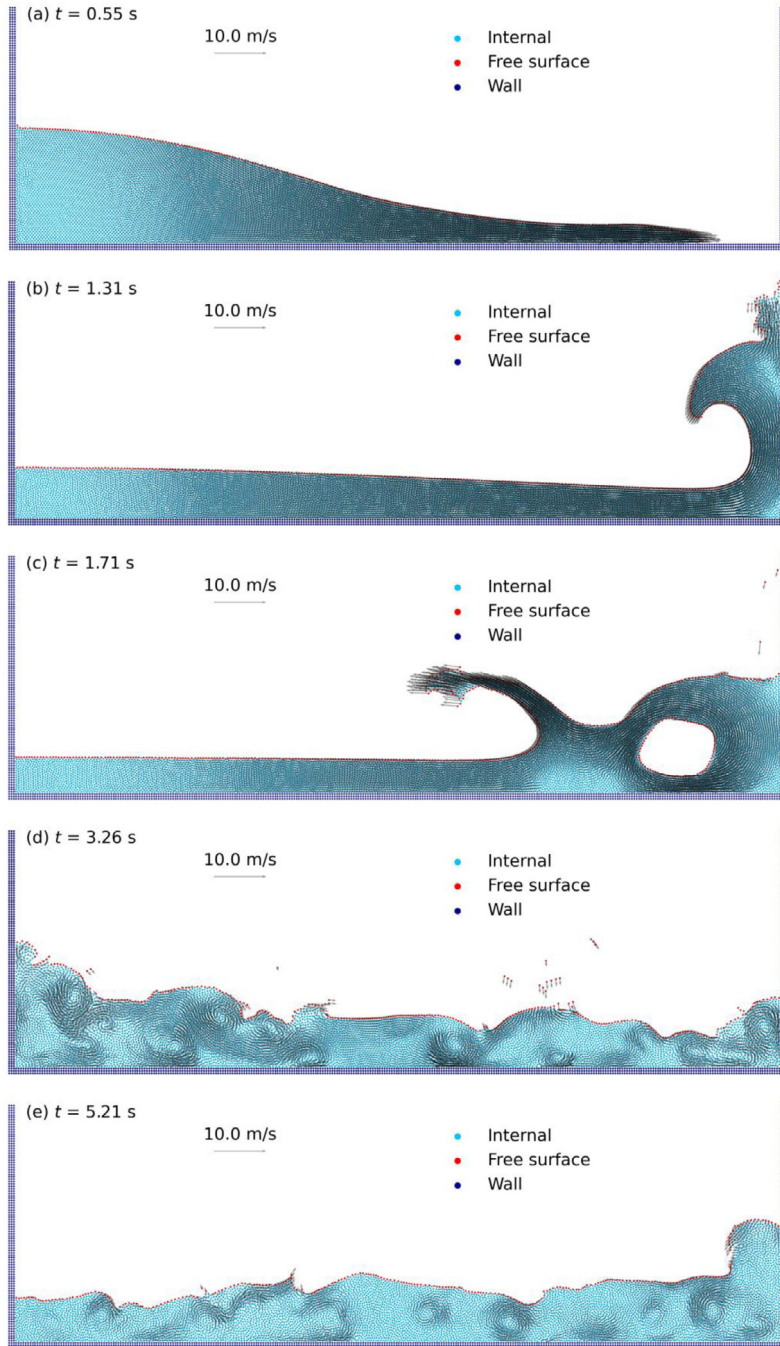


Fig. 28. Simulated velocity distributions and free-surface detection using the proposed conditions at several typical moments in the dam break flow.

model produced additional velocity diffusion, the existing conditions still could not provide stable results in this problem. On the other hand, when the proposed conditions were applied, the maximum velocity was usually less than 5 m/s (Fig. 32), and the velocity distributions were much smoother than the ones using the existing conditions (see Fig. 33). Therefore, the proposed conditions produced much more stable results than the existing ones in the violent free-surface flow.

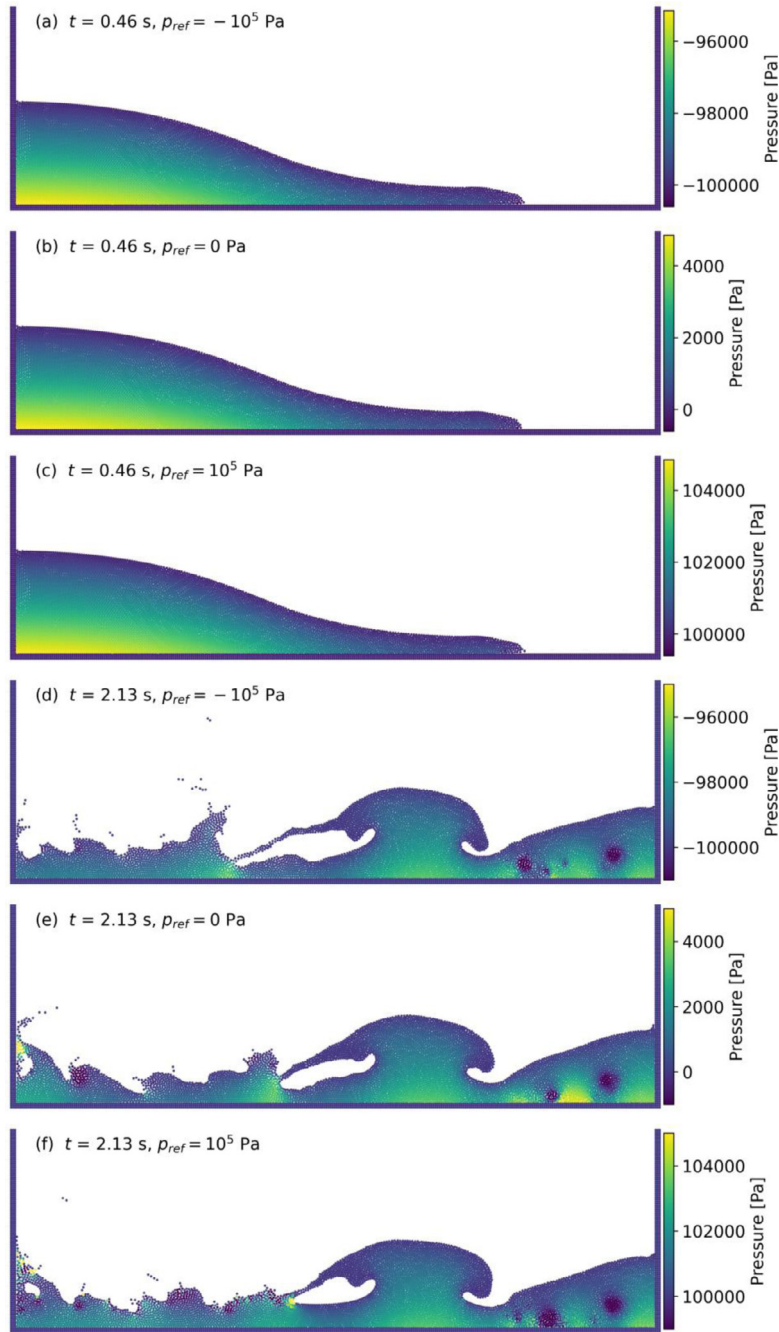
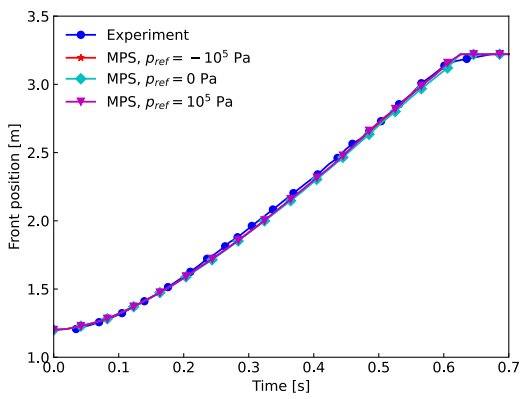


Fig. 29. Snapshots of pressure distributions simulated using different reference pressure at free surfaces (p_{ref}) in the dam break flow.

Second, typical snapshots simulated using the proposed conditions are presented. As shown in Fig. 34, after the downward flow impacted to the bottom wall at $t = 0.4$ s, it would spread quickly and form two vortices. At $t = 0.6$ s, the two vortices increased further. At $t = 0.9$ s, the right vortex above the barrier rose further and become physically unstable. The splash took place at $t = 1.5$ s due to the vortex rise, and a clear vortex was formed in the left area. Then, from $t = 2.0$ s to 4.0 s, various splash, fragmentation, and coalescence of free surfaces were



(a) Front positions

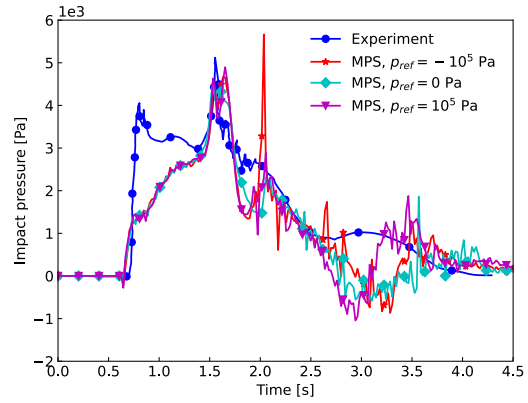
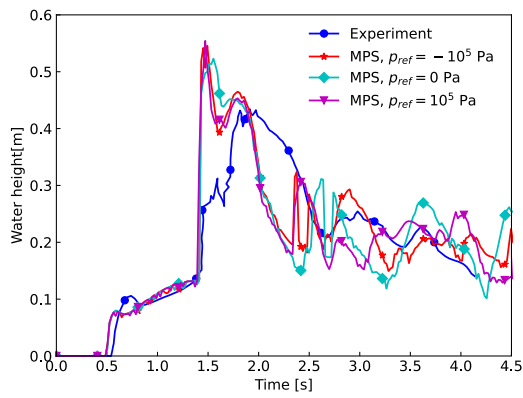
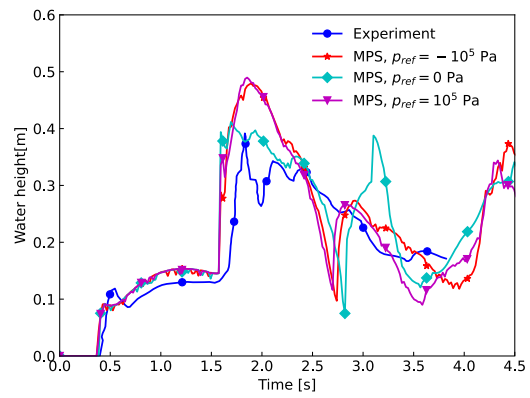
(b) Pressure at point P_1 (c) Water height at position H_1 (d) Water height at position H_2

Fig. 30. Histories of some quantitative results simulated using different reference pressure at free surfaces (p_{ref}) in the dam break flow.

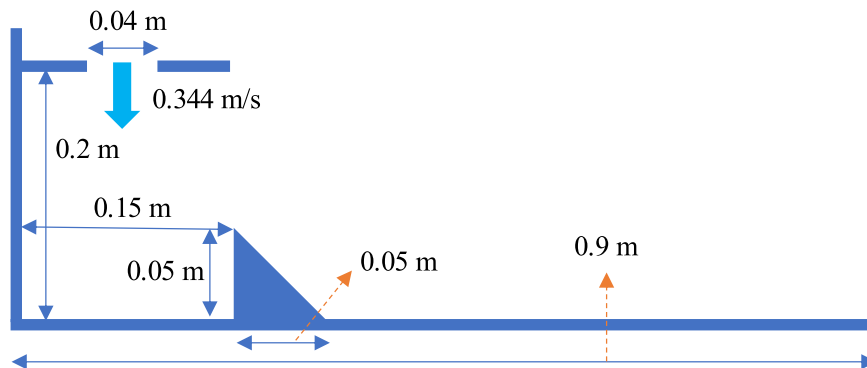


Fig. 31. Sketch of the initial configuration in the inflow impact problem.

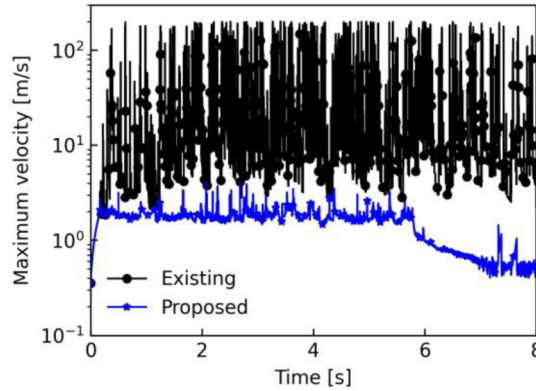


Fig. 32. Comparison of maximum velocity simulated by the existing and proposed conditions in the inflow impact problem.

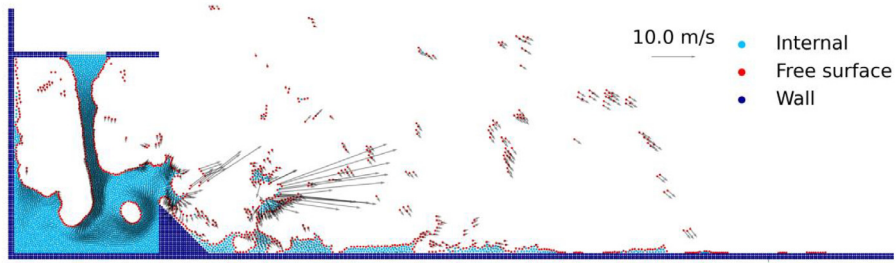
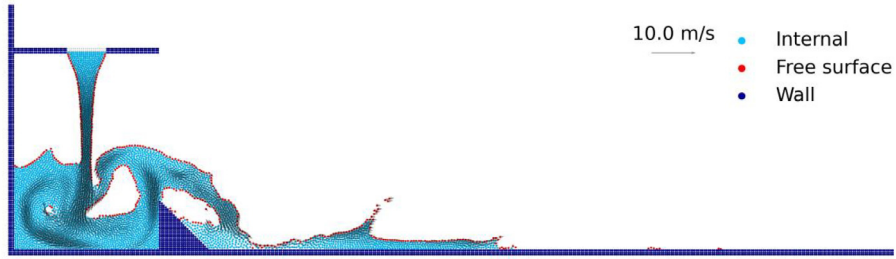
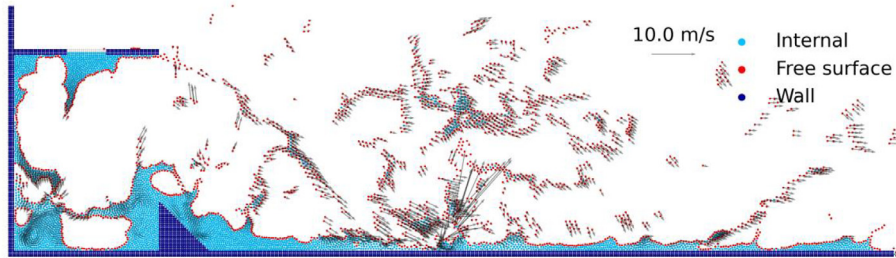
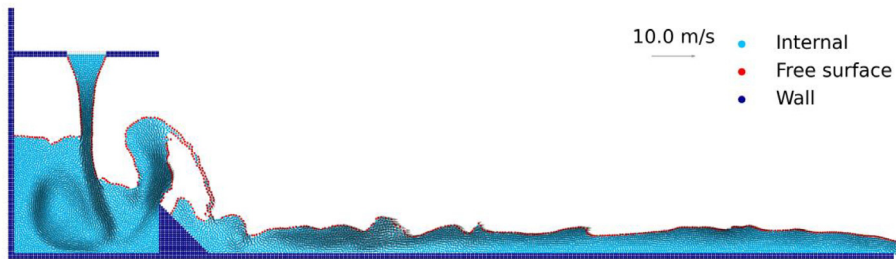
observed due to the strong interaction between vortices and free surfaces near the barrier. The simulated velocity distributions were all smooth in Fig. 34. Thus, the proposed conditions could produce the stable and reasonable results for the violent problem.

6. Concluding remarks

This paper investigated the numerical instability near free surfaces when the consistent high-order schemes were applied in a semi-implicit particle method. A semi-analytical error-analysis method was proposed to clarify the mechanism for the error accumulation due to the biased neighbor support. Because the center fluctuation was the largest in the BEAP, the discretization models were decomposed into the CVCs and NVCs. The mechanism how the CVCs could cause error accumulation was clarified by a theoretical analysis. Then, the influence of the NVCs on error accumulation was considered from numerical tests. Meanwhile, the influence of the negative Laplacian-model coefficients on PPE convergence was discussed analytically and numerically. Novel conditions for free surface detection were proposed based on the error analysis. The performance of the proposed indices was tested by various numerical tests and examples. The main conclusions are as follows.

- (1) When an internal particle took a severely biased neighbor support, its pressure and velocity would excite each other via a positive feedback loop of CVCs, thus resulting in error accumulation. If the internal particle was detected as a free surface one, the positive feedback loop was broken, and thus the error accumulation stopped. Therefore, detecting the particle as a free-surface particle could effectively suppress error accumulation.
- (2) The biased CVCs always had a tendency to result in error accumulation. The NVCs tended to attenuate the error accumulation. When the bias of neighbor support was just severe enough to make the CVC effect exceed the NVC effect, error accumulation would occur. The indices R_i was proposed to measure the error-accumulation risk. The larger the R_i became, the higher the risk was.
- (3) The negative Laplacian-model coefficients corresponded to the unphysical negative diffusion. The inclusion of negative coefficients into the PPE system could result in fluctuations of the convergent iteration number and increase the required iteration number. The indices c_i , and L_i were proposed to measure the negative-coefficient contribution.
- (4) Four quantitative indices, $\|b_i\|$, c_i , R_i , and L_i , were proposed to measure the bias/incompleteness of neighbor supports. After the indices were considered into the free-surface-detection conditions, the simulation stability was substantially enhanced compared to the existing conditions. The proposed conditions were found to provide stable simulations in the various cases with positive/negative pressure, violent deformation, and topological changes of free surfaces. The 3D extension of the method would be straightforward because no geometric information was involved in the proposed indices.

The developed method only depended on the variable difference, and thus arbitrary pressure could be imposed at free surfaces. The error-analysis method and the proposed indices effectively suppressed the spurious currents due

(a) Existing conditions at $t = 1.24$ s(b) Proposed conditions at $t = 1.24$ s(c) Existing conditions at $t = 3.75$ s(d) Proposed conditions at $t = 3.75$ s**Fig. 33.** Comparison of simulated velocity distribution using the existing and proposed conditions in the inflow impact problem.

to fast error accumulation. Thus, they are of great significance to the application of the consistent schemes. The extensions of the proposed methodology to 3D and to the explicit algorithm will be our future work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

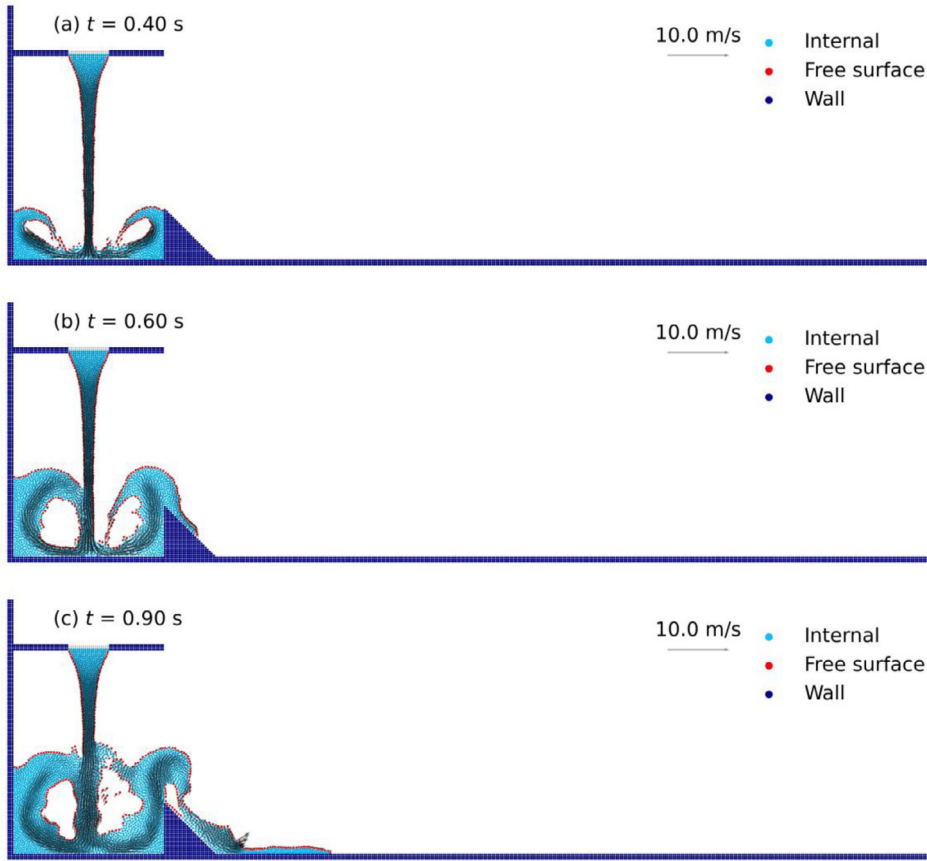


Fig. 34. Snapshots of the simulated velocity distributions using the proposed condition at different moments in the inflow impact problem.

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Appendix. Error accumulation via center-variable components

As mentioned in the main body, error accumulates via the velocity divergence, PPE, and pressure gradient. The error accumulation via the CVCs is derived herein.

Because velocity divergence is the main part of the PPE source, the following PPE is adopted for analysis:

$$\frac{1}{\rho} \langle \nabla^2 p \rangle_i^{k+1} = \frac{1}{\Delta t} \langle \nabla \cdot \mathbf{u} \rangle_i^k, \quad (\text{A.1})$$

where k indicates the time-step label. From the model decomposition [Eqs. (38) and (42)], we have the following identity:

$$\frac{1}{\rho} L_j^{k+1} + \frac{1}{\rho} L_i^{k+1} = \frac{1}{\Delta t} D_j^k + \frac{1}{\Delta t} D_i^k. \quad (\text{A.2})$$

For example, D_i^k and L_i^{k+1} are defined in Eqs. (40) and (44), respectively. Because D_j^k and L_j^{k+1} are involved in Eq. (A.2), a global solution for the error transfer is still necessary. Fortunately, the main conclusion of Section 3.2 is that D_i^k and L_i^{k+1} are the dominant reasons for error accumulation. Thus, the NVCs are neglected temporarily,

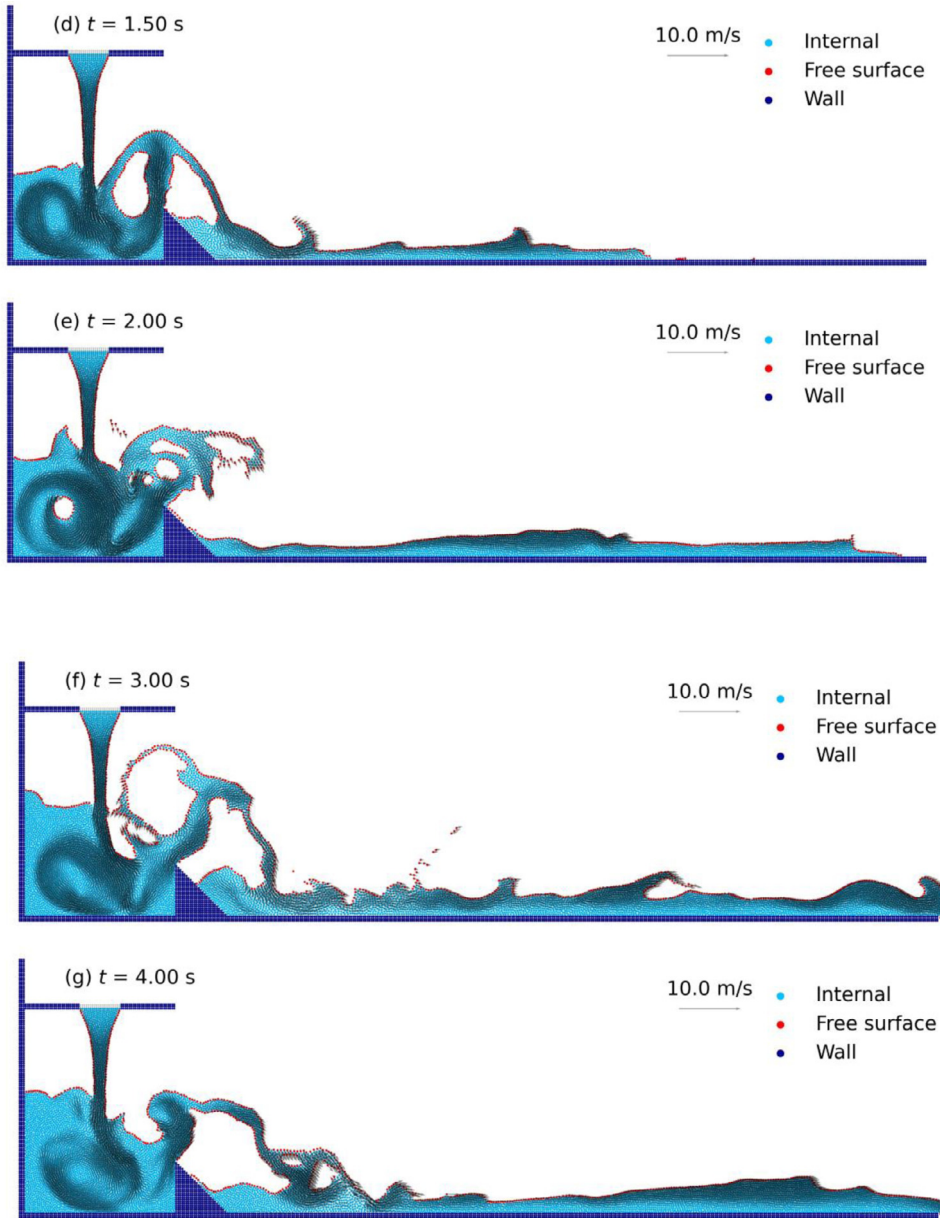


Fig. 34. (continued).

and their influence is discussed in Section 3.4. In this situation, Eq. (A.2) can be simplified to

$$\frac{1}{\rho} L_i^{k+1} \approx \frac{1}{\Delta t} D_i^k. \quad (\text{A.3})$$

In Eq. (A.3), a global solution becomes unnecessary, and the accumulative fluctuation is transferred only via each particle separately.

The velocity update after the pressure-gradient calculation is

$$\mathbf{u}_i^{k+1} = \mathbf{u}_i^k - \frac{\Delta t}{\rho} \langle \nabla p \rangle_i^{k+1}. \quad (\text{A.4})$$

Based on the decomposition in Eq. (32), it is written as:

$$\mathbf{u}_i^{k+1} = \mathbf{u}_i^k - \frac{\Delta t}{\rho} \mathbf{G}_j^{k+1} - \frac{\Delta t}{\rho} \mathbf{G}_i^{k+1}. \quad (\text{A.5})$$

As discussed in Section 3.2, $\mathbf{G}_i^{k+1}$ is dominant compared to $\mathbf{G}_j^{k+1}$ during error accumulation. Thus, Eq. (A.5) can be further simplified to

$$\mathbf{u}_i^{k+1} \approx \mathbf{u}_i^k - \frac{\Delta t}{\rho} \mathbf{G}_i^{k+1}. \quad (\text{A.6})$$

Finally, the simplified governing equations are Eqs. (A.3) and (A.6).

Next, we analyze the mechanism of error accumulation. Let $\mathbf{m}$ be the unit vector of $\mathbf{b}_i$, namely $\|\mathbf{b}_i\| = \mathbf{m} \cdot \mathbf{b}_i$. As mentioned in Section 3.2, the velocity only accumulates in the direction of $\mathbf{b}_i$ or $\mathbf{m}$. Let us therefore assume that particle i is suffering from error accumulation and that the initial velocity is

$$\mathbf{u}_i^k = a \mathbf{m}, \quad (\text{A.7})$$

where the coefficient a can be positive or negative. Based on Eq. (40), D_i^k in the simplified PPE [Eq. (A.3)] can be calculated as

$$D_i^k = -\frac{\mathbf{u}_i^k}{r_e} \cdot \mathbf{b}_i = -\frac{a}{r_e} \mathbf{m} \cdot \mathbf{b}_i = -\frac{a}{r_e} \|\mathbf{b}_i\|. \quad (\text{A.8})$$

Inserting the definition of L_i^k in Eq. (44) and the calculated D_i^k into Eq. (A.3), gives

$$-\frac{1}{\rho} \frac{p_i^{k+1}}{l_0 r_e} c_i \approx -\frac{1}{\Delta t} \frac{a}{r_e} \|\mathbf{b}_i\|. \quad (\text{A.9})$$

From Eq. (A.9), the pressure at the next time step can thus be approximated as

$$p_i^{k+1} \approx a l_0 \frac{\rho}{\Delta t} \frac{\|\mathbf{b}_i\|}{c_i}, \quad (\text{A.10})$$

The calculated p_i^{k+1} from Eq. (A.10) can then be used to update the velocity based on Eq. (A.6). Thus, inserting p_i^{k+1} and $\mathbf{G}_i^{k+1}$ [in Eq. (35)] into Eq. (A.6), we obtain

$$\mathbf{u}_i^{k+1} \approx \mathbf{u}_i^k + \frac{\Delta t}{\rho} \frac{p_i^{k+1}}{r_e} \mathbf{b}_i = a \mathbf{m} + a \frac{l_0}{r_e} \frac{\|\mathbf{b}_i\|}{c_i} \mathbf{b}_i = a \mathbf{m} \left(1 + \frac{l_0}{r_e} \frac{\|\mathbf{b}_i\|^2}{c_i} \right), \quad (\text{A.11})$$

where the identity $\mathbf{b}_i = \|\mathbf{b}_i\| \mathbf{m}$ is used. Based on Eqs. (A.7) and (A.11), the growth rate of the accumulated velocity after one time step can be calculated as follows:

$$\mathbf{u}_i^{k+1} = \left(1 + \frac{l_0}{r_e} \frac{\|\mathbf{b}_i\|^2}{c_i} \right) \mathbf{u}_i^k, \quad (\text{A.12})$$

where $\mathbf{u}_i^{k+1}$ and $\mathbf{u}_i^k$ are in the same direction, and the time step Δt is not involved. The factor l_0/r_e arises from the normalization methods used in defining $\mathbf{b}_i$ and c_i in Eqs. (36) and (45), respectively.

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